

Guarded block transfers for polytope distance: linear convergence with explicit constants and gap-certified screening for SVM training

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Abstract

Training a hard-margin support vector machine is the distance problem between two convex hulls, which the Triangle Algorithm solves with a certified duality gap. We study *guarded block transfers*: k Mitchell–Demjanov–Malozemov transfers per scan are aggregated into one exact line search, guarded by a comparison with the single best transfer, and replaced by an away step when the leading transfer is capacity-clipped. On the Minkowski-difference polytope we prove global linear convergence with explicit constants and no combinatorial swap-step bounds, and a finite activation bound for gap-safe screening. We then ask when aggregation pays. The gain of a block over the single transfer at the same iterate is exactly Q/κ in two quantities the guard already computes; under strict complementarity every drop step occurs before an explicit identification iteration; and a fitted work model gives the block size that minimises the per-iteration term, $\sqrt{(a/b)(1-\eta)/\eta}$, which the solver sets adaptively from κ . Isolated measurements on five LIBSVM problems confirm the prediction: aggregation pays only where per-iteration work rather than column computation dominates, by a factor bounded by the column share. The solver ties LIBSVM on sparse-support hard margins, beats BPCG and kernel SMO on kernel L_2 margins, and is one to two orders of magnitude slower than LIBLINEAR on dense-support L_2 margins. We report both sides.

Keywords: polytope distance; Frank–Wolfe methods; Triangle Algorithm; support vector machines; safe screening; linear convergence

1 Introduction

Training a linear hard-margin SVM is equivalent to finding the closest points of the two classes’ convex hulls: the optimal separator is the perpendicular bisector of the witness pair and the margin is half the hull distance (Bennett and Bredensteiner, 2000). Kalantari’s Triangle Algorithm (TA) exploits this *distance duality* directly (Kalantari, 2015): TA I decides whether the hulls intersect, and TA II computes their distance while maintaining an iterate pair (p, q) and a certified interval $[LB, UB]$ from parallel supporting hyperplanes. The enhanced TA (ETA) of Gupta and Kalantari (2026) adds dot-product caching and search heuristics.

The workhorse step of such solvers is the *weight transfer* from the worst active vertex to the best candidate, the Mitchell–Demjanov–Malozemov (MDM) step (Mitchell et al., 1974), which is the pairwise Frank–Wolfe (FW) step of (Lacoste-Julien and Jaggi, 2015) on the relevant polytope. Each step costs an $O(n)$ scan and yields one transfer; we ask how much of the scan can be reused across several transfers without weakening the guarantee.

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Contributions.

1. **Guarded block transfers** (§3): pair the top- k receivers with the worst- k active donors, take one exact line search along the aggregated direction, *guard* it by an $O(k^2)$ comparison with the single MDM step, and fall back to an away step on capacity-clipped iterations.
2. **Linear convergence with explicit constants** (Theorem 6): on $Z = \text{conv}(V) - \text{conv}(W)$ the guarded block algorithm inherits the pyramidal-width rate of pairwise FW up to a constant, and the away fallback replaces the combinatorial swap-step bound by linear drop-step counting (cf. (Tsuji et al., 2022)).
3. **Screening activation with explicit constants** (Theorem 11): the TA certificate is the FW gap divided by the distance (Lemma 10), so gap-safe screening (Ndiaye et al., 2017) applies, and we bound the iteration by which it has removed every zero-weight point off the optimal faces.
4. **A regime study on real data** (§6): a heuristic pyramidal-width estimate ($\sim 1/\sqrt{n}$) suggests $O(n^2)$ -type cost on dense supports, and two batteries on five LIBSVM datasets show the boundary from both sides.
5. **When aggregation pays** (§7, new in this version): the exact gain ratio of a block over the single MDM step at the same iterate, $Q(2 - \kappa)$ or Q/κ in two observable numbers (Proposition 12); an identification theorem showing that under strict complementarity all drop steps occur before an explicit iteration and none afterwards (Theorem 14); a conditional work bound with the block size that minimises it (Corollary 15); isolated per-step measurements of every quantity in the model on five real cells; and an adaptive block size driven by the measured κ , tested prospectively.

Code, proofs, numerical checks and experiment scripts are in the public repository ([\[TODO: URL\]](#)).

2 Setting

Let $V = \{v_1, \dots, v_n\}$, $W = \{w_1, \dots, w_m\} \subset \mathbb{R}^d$. The polytope distance problem is

$$\min_{p \in \text{conv}(V), q \in \text{conv}(W)} \frac{1}{2} \|p - q\|^2. \quad (1)$$

Writing $z = p - q$ and $Z := \text{conv}(V) - \text{conv}(W) = \text{conv}\{v_i - w_j\}$, (1) is the minimum-norm-point problem $\min_{z \in Z} F(z)$, $F(z) = \frac{1}{2} \|z\|^2$, whose objective is 1-strongly convex and 1-smooth. Let z^* be the minimiser, $\delta^* = \|z^*\|$, $h(z) = F(z) - F(z^*)$, $M = \text{diam}(Z)$ and $\delta = \text{PW}(A) > 0$ the pyramidal width (Lacoste-Julien and Jaggi, 2015) of the atom set $A = \{v_i - w_j\}$ (a property of the atom set, not of Z alone).

The algorithm keeps simplex weights α on V , β on W (iterate z_t after t iterations), with $p = \sum \alpha_i v_i$, $q = \sum \beta_j w_j$; the induced product representation $\lambda_{ij} = \alpha_i \beta_j$ of z over the atoms $v_i - w_j$ has support $\text{supp } \alpha \times \text{supp } \beta$. With $g = -\nabla F(z) = q - p$, the *side scores* are $s_i^V = \langle g, v_i \rangle$ and $s_j^W = \langle p - q, w_j \rangle$, maintained incrementally from cached arrays.

Certificate. For any iterate, $\text{UB} = \|z\|$ and $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\| = (\min_i \langle z, v_i \rangle - \max_j \langle z, w_j \rangle) / \text{UB}$ satisfy $\text{LB} \leq \delta^* \leq \text{UB}$; the TA stops when the certified relative gap $(\text{UB} - \text{LB}) / \text{UB}$ is at most ε . The L_2 soft margin with regularisation parameter C , the ν -SVM and the kernel variants reduce to (1) exactly (augmented coordinates that turn the Gram matrix K into $K + I/C$; reduced convex hulls; feature-space images), see Section 5.

Algorithm 1: One iteration of the guarded block-transfer step (side V ; W symmetric)

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1 top- $k$  receivers by  $s^V$ ; worst- $k$  donors in  $\text{supp } \alpha$ ; pair best-with-worst:  $(r_j, u_j)$ ,  $j \leq k'$ 
  ( $k' \leq k$  pairs, limited by the support size); pair 1 is the MDM pair  $(r^*, u^*)$ 
2  $d_j \leftarrow v_{r_j} - v_{u_j}$ ;  $\text{gap}_j \leftarrow s_{r_j}^V - s_{u_j}^V$ ; discard pairs with  $\text{gap}_j \leq 0$  and receivers that are
  donors (if  $r^*$  is a donor it leaves the donor list, so pair 1 survives);
   $\gamma_j \leftarrow \min(\text{gap}_j / \|d_j\|^2, c_j)$  //  $c_j = \alpha_{u_j}$ : donor capacity
3 if pair 1 uncapped ( $\gamma_1 = \text{gap}_1 / \|d_1\|^2$ ) [case (a)] then
4    $\Delta_B \leftarrow$  gain of exact line search  $\theta \in [0, 1]$  along  $d = \sum_j \gamma_j d_j$ ;  $\Delta_1 \leftarrow$  gain of the
   MDM step
5   take the step with the larger gain // guard,  $O(k'^2)$ 
6 else
7   if the away step on  $u^*$  is boundary-clipped [case (b)] then
8     take it // drop step:  $\alpha_{u^*} \rightarrow 0$ 
9   else
10    take the best of {block, MDM, away( $u^*$ ), toward( $r^*$ )}
11  end
12 end

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Gaps. $g^{\text{FW}} = \max_{u \in Z} \langle g, u \rangle - \langle g, z \rangle$, $g^{\text{A}} = \langle g, z \rangle - \min_{u \in \text{supp } \lambda} \langle g, u \rangle$, $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}}$. Because $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$ is separable and the support is a product set, each gap splits into a V -side and a W -side term computable from the score arrays in $O(n+m)$ (Lemma 1), written $g_V^{\text{PW}}, g_W^{\text{PW}}$ (likewise for FW and A), with g_{side} the term of the side being stepped; a transfer $v_r - v_u$ on the V side is a pairwise direction of Z (Lemma 2).

Lemma 1 (Separability). *With the product representation, $g^{\text{PW}} = g_V^{\text{PW}} + g_W^{\text{PW}}$ where $g_V^{\text{PW}} = \max_i s_i^V - \min_{i \in \text{supp } \alpha} s_i^V \geq 0$ and g_W^{PW} likewise; the same holds for g^{FW} and g^{A} .*

Proof. $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$ is separable and the support is a product set, so the max over atoms and the min over the support separate. Each side term is nonnegative: the FW term is the maximum score minus the weighted mean of the support scores, the away term is that mean minus the support minimum, and a weighted mean lies between the minimum and the maximum. $\square$

Lemma 2 (Side transfers are Z -pairwise directions). *For $\alpha_u > 0$ and any $b \in \text{supp } \beta$, $v_r - v_u = (v_r - w_b) - (v_u - w_b)$ is a difference of an atom of Z and an atom in the support, and the transfer $\gamma \leq \alpha_u$ realises $z \leftarrow z + \gamma(v_r - v_u)$ inside Z . The side capacity $\alpha_u = \sum_b \alpha_u \beta_b$ is the total weight of the product atoms $v_u - w_b$, $b \in \text{supp } \beta$: the side transfer is the sum of the atom-pair transfers of masses $\gamma \beta_b$ from $v_u - w_b$ to $v_r - w_b$, and for $r \neq u$ this weight update is feasible exactly when $0 \leq \gamma \leq \alpha_u$; hence every such transfer remains in Z .*

3 Guarded block transfers

Algorithm 1 amortises one $O(n)$ scan over up to k transfers. Only the scan is amortised: a new receiver still costs one Gram or kernel column, $O(nd)$. Every candidate gain is available from cached quantities at a cost independent of n : with $S = \langle g, d \rangle = \sum_j \gamma_j \text{gap}_j$ and $D = \|d\|^2 = \gamma^\top \Gamma \gamma$, Γ the $k' \times k'$ Gram matrix of the pair directions ($O(k'^2)$ cached dot products), the block gain is $\theta S - \theta^2 D / 2$ at $\theta = \min(S/D, 1)$; the MDM, away and toward gains are the usual one-dimensional expressions. The *gain* of a step is the decrease $h(z_t) - h(z_{t+1})$. The cache update for a block is one matrix–vector product with the net per-index coefficients; columns are computed on first use and cached, so support columns cost nothing after their first appearance.

Feasibility is immediate, since donors lose at most $\theta\gamma_j \leq c_j$. We call an iteration *case (a)* if pair 1 is uncapped and *case (b)* otherwise; the away step of line 6, which drives α_{u^*} to zero, is a *drop* step (a capped block pair may also zero a donor, which only helps the counts below).

Safe screening. With $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$ and $v_{\min} = \arg \min_i \langle z, v_i \rangle$, a zero-weight point with $\langle z, v_i \rangle - \langle z, v_{\min} \rangle > r\|v_i - v_{\min}\|$ is outside the optimal support and is removed (symmetrically on W with the max). Because the optimum's support survives, the reduced problem has the same optimum and all subsequent lower bounds remain valid.

4 Analysis

Conventions. $M > 0$ (a singleton Z is solved at initialisation), and the algorithm stops with “intersect” when $\text{UB} = \|z\| = 0$, before any division by UB . Lemma 3 and Algorithm 1 presuppose $k' \geq 1$: a side with no positive-gap pair is skipped, and on a non-drop iteration the side with the larger pairwise gap has one. An away direction whose donor has weight one is unavailable (the iterate is that vertex and the away gap of that side is zero). In Proposition 9, $\eta \geq 0$.

Lemma 3 (Block gain). *Let $b_j = \gamma_j\|d_j\|$ over the k' retained pairs, all with $\text{gap}_j > 0$. Then $S \geq \sum_j b_j^2$, $D \leq k'S$, and the realised block gain satisfies $\Delta_B \geq S/(2k') \geq \Delta_1/(2k')$, Δ_1 the MDM gain.*

Proof of Lemma 3. Since $\gamma_j \leq \text{gap}_j/\|d_j\|$, we have $\gamma_j\text{gap}_j \geq \gamma_j^2\|d_j\|^2 = b_j^2$. Consequently, $S = \sum_j \gamma_j\text{gap}_j \geq \sum_j b_j^2$. By the triangle inequality, $\|d\| \leq \sum_j \gamma_j\|d_j\| = \sum_j b_j$, and by Cauchy–Schwarz, $(\sum_j b_j)^2 \leq k' \sum_j b_j^2 \leq k'S$; hence $D \leq k'S$. Exact line search on the 1-smooth F gives $\Delta_B = S^2/(2D) \geq S/(2k')$ in the interior case and $\Delta_B = S - D/2 \geq S/2$ at the boundary $\theta = 1$, where $D \leq S$. Finally, $S \geq \gamma_1\text{gap}_1 \geq \Delta_1$, since all terms are nonnegative and $\Delta_1 = \gamma_1\text{gap}_1 - \gamma_1^2\|d_1\|^2/2$. $\square$

With the guard, every case-(a) iteration gains $\max(\Delta_B, \Delta_1) \geq \Delta_1$: the guarded step never makes less progress than pairwise FW.

Lemma 4 (Good-step progress). *If the iteration works on the side with the larger pairwise gap, then on any non-drop iteration gain $\geq \min((g^{\text{PW}})^2/(32M^2), g^{\text{PW}}/8)$.*

Proof of Lemma 4. Let $g_S = \max(g_V^{\text{PW}}, g_W^{\text{PW}}) \geq g^{\text{PW}}/2$. Every direction below is a difference of two points of one class hull, so its norm is at most $\text{diam}(\text{conv } V) \leq M$. For a feasible direction with directional derivative $\sigma > 0$, squared norm D and maximal step $\bar{\theta}$, exact line search on a 1-smooth function gives gain $\geq \min(\sigma^2/(2D), \bar{\theta}\sigma/2)$ (interior optimum, or the clipped step evaluated with $\bar{\theta}D \leq \sigma$). In case (a), the guard's gain is at least the MDM gain, whose derivative is g_S and whose interior optimum is attained, yielding gain $\geq g_S^2/(2M^2) \geq (g^{\text{PW}})^2/(8M^2)$. In case (b), non-drop, $g_S = g_{\text{side}}^{\text{FW}} + g_{\text{side}}^{\text{A}}$, so one term is at least $g_S/2 \geq g^{\text{PW}}/4$. If it is the FW term, the toward step (maximal step 1) gives gain $\geq \min((g_{\text{side}}^{\text{FW}})^2/(2M^2), g_{\text{side}}^{\text{FW}}/2) \geq \min((g^{\text{PW}})^2/(32M^2), g^{\text{PW}}/8)$; if it is the away term, the away line search is interior (non-drop) and gain $\geq (g_{\text{side}}^{\text{A}})^2/(2M^2) \geq (g^{\text{PW}})^2/(32M^2)$. The algorithm takes the best available candidate. $\square$

Lemma 5 (Geometric strong convexity; (Lacoste-Julien and Jaggi, 2015, Thm. 6) with $\mu = 1$). *For every $z \in Z$ with a representation whose support contains an away atom, $g^{\text{PW}}(z) \geq \delta\sqrt{2h(z)}$ and $g^{\text{PW}}(z) \geq g^{\text{FW}}(z) \geq h(z)$.*

Proof. This is Theorem 6 of (Lacoste-Julien and Jaggi, 2015) for a 1-strongly convex objective; we include the short argument. If $z = z^*$ both inequalities are immediate ($h = 0$ and all gaps are

nonnegative). Otherwise let $e = (z^* - z)/\|z^* - z\|$; the pyramidal-width inequality in the proof of (Lacoste-Julien and Jaggi, 2015, Thm. 6) (Definition 3 there, applied to the direction g together with $\langle g, e \rangle \leq \|g\|$) gives, for the FW atom s and the away atom v , $\langle g, s - v \rangle \geq \delta \langle g, e \rangle$. By 1-strong convexity, $\langle g, z^* - z \rangle \geq h + \frac{1}{2} \|z^* - z\|^2$, so with $\varrho = \|z^* - z\|$, $g^{\text{PW}} \geq \delta(h/\varrho + \varrho/2) \geq \delta\sqrt{2h}$ by AM-GM. Finally $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}} \geq g^{\text{FW}} = \max_u \langle g, u - z \rangle \geq \langle g, z^* - z \rangle \geq h$ by convexity. $\square$

Theorem 6 (Global linear convergence). *Suppose the guarded block-transfer algorithm with away fallback is run for one side per iteration (the side with the larger pairwise gap), using any $k \geq 1$, from any vertex pair (initial support $s_0 = 2$). Let $h_t = h(z_t)$. Then every non-drop iteration satisfies $h_{t+1} \leq (1 - \rho)h_t$ with*

$$\rho = \frac{1}{16} (\delta/M)^2.$$

For any horizon T , among the first T iterations the number of drop steps is at most $kT_{\text{good}} + s_0$, where T_{good} is the number of non-drop iterations; consequently, $h_T \leq h_0 (1 - \rho)^{(T-s_0)/(k+1)}$. Under the two-sided schedule of the implementation (larger-gap side first, second side skipped after a drop) every non-drop iteration has the same contraction, the drop count is at most $2kT_{\text{good}} + s_0$, and $h_T \leq h_0 (1 - \rho)^{(T-s_0)/(2k+1)}$ (Lemma 7).

The rate combines Lemma 4 with $g^{\text{PW}} \geq \delta\sqrt{2h}$ and $g^{\text{PW}} \geq g^{\text{FW}} \geq h$ (Lemma 5). Drop counting is the away-step argument: each drop removes one support index and non-drop steps add at most k . No swap-step bound is needed: a case-(b) iteration either drops (counted above) or takes the best of four candidates, of which the toward or away step alone gains Lemma 4's amount, whichever wins. Table 2 used the earlier unconditional V -then- W order (Table 3).

Proof of Theorem 6. By Lemmas 4 and 5 a non-drop iteration gains at least

$$\min(2\delta^2 h / (32M^2), h/8) = (\delta/M)^2 h / 16$$

using $\delta \leq M$; hence $h_{t+1} \leq (1 - \rho)h_t$. Drop counting: non-drop iterations add at most k support indices, drop iterations add none and remove at least one, and the support size stays ≥ 2 from s_0 ; so the number of removals, hence of drops, is at most $kT_{\text{good}} + s_0$, so that

$$T_{\text{good}} \geq (T - s_0)/(k + 1).$$

All steps, drops included, are descent steps, so h is monotone. $\square$

Lemma 7 (Two-sided schedule). *Let every iteration compute both side gaps, take the guarded step of Algorithm 1 on the side with the larger pairwise gap first, and take the other side's guarded step only if the first step was not a drop; an iteration counts as a drop iff its first step is a drop. Then every non-drop iteration gains at least $\min((g^{\text{PW}})^2 / (32M^2), g^{\text{PW}}/8)$ and the number of drops is at most $2kT_{\text{good}} + s_0$; hence Theorem 6 holds for this schedule with the same ρ and exponent $(T - s_0)/(2k + 1)$.*

Proof. On a non-drop iteration the first step is exactly the step analysed in Lemma 4, so it gains at least the stated amount; the second step is an exact line search along a feasible descent direction (or is skipped), so it cannot increase h . Drop counting: a drop iteration takes a single away step that removes one support index and adds none, because the second side is skipped; a non-drop iteration takes two guarded steps and adds at most k indices each. Removals, hence drops, therefore number at most $2kT_{\text{good}} + s_0$. $\square$

Remark 8. *The skip is what makes the count work. If the second side always steps, an iteration whose analysed side drops can add up to k indices on the other side, and the count becomes $T_{\text{drop}} \leq 2kT_{\text{good}} + kT_{\text{drop}} + s_0$, which is vacuous for $k \geq 1$.*

Proposition 9 (Near-orthogonal speed-up). *If all k' pairs are uncapped and $|\langle d_i, d_j \rangle| \leq \eta \|d_i\| \|d_j\|$ for $i \neq j$, then the block gain is at least $(\sum_j \Delta_j)/(1 + \eta(k' - 1))$, where Δ_j is pair j 's individual gain.*

Proof of Proposition 9. $D = \sum_{i,j} \gamma_i \gamma_j \langle d_i, d_j \rangle \leq \sum_j b_j^2 + \eta \sum_{i \neq j} b_i b_j \leq (1 + \eta(k' - 1)) \sum_j b_j^2 \leq (1 + \eta(k' - 1))S$ by Cauchy–Schwarz on the cross terms. If $D \geq S$ the line search is interior and $\Delta_B = S^2/(2D) \geq S/(2(1 + \eta(k' - 1)))$; if $D < S$ then $\theta = 1$ and $\Delta_B = S - D/2 > S/2$, which is larger. For uncapped pairs $S = \sum_j 2\Delta_j$. $\square$

As $\eta \rightarrow 0$ one scan realises the sum of all k pairwise gains.

Lemma 10 (The certified gap is the FW gap). *At any full scan, with LB evaluated at the current $z \neq 0$, $UB - LB = g^{\text{FW}}(z)/UB$ (a retained best lower bound gives $\leq$), and $g^{\text{FW}}(z) \leq 2h + R\sqrt{2h}$ with $R = \max_{u \in Z} \|u\|$, the radius of Z about the origin.*

Proof of Lemma 10. $LB = \min_{u \in Z} \langle z, u \rangle / \|z\|$ by definition of the supporting hyperplane bound and $UB = \|z\|$; subtracting, $UB - LB = (\langle z, z \rangle - \min_u \langle z, u \rangle) / \|z\| = g^{\text{FW}}/UB$. At the optimum $0 = \langle z^*, z^* \rangle - \min_u \langle z^*, u \rangle$; subtracting from $g^{\text{FW}}(z)$, $g^{\text{FW}}(z) = (\|z\|^2 - \|z^*\|^2) + (\min_u \langle z^*, u \rangle - \min_u \langle z, u \rangle) \leq 2h + \max_u \langle z^* - z, u \rangle \leq 2h + R\|z - z^*\| \leq 2h + R\sqrt{2h}$ by 1-strong convexity. $\square$

Theorem 11 (Screening activation). *Assume $\delta^* > 0$. Let $F_V = \{i : \langle z^*, v_i \rangle = \min_j \langle z^*, v_j \rangle\}$ and, with the max, F_W be the optimal faces: the vertices of zero score margin, which contain the optimal supports (possibly strictly). Let $\tau_i = \langle z^*, v_i \rangle - \min_j \langle z^*, v_j \rangle$ be the score margin of v_i (and symmetrically on W), with $\tau > 0$ the minimal nonzero margin. Finally, let L be the larger class diameter.*

*A zero-weight point with margin τ_i is screened as soon as $r < \tau_i/(3L)$. Consequently, once $h_t < H := \tau^4 \delta^{*2}/(2592 R^4 L^4)$, every zero-weight point of V outside F_V and of W outside F_W is removed permanently at the next screening round. Under Theorem 6 this holds for every iteration $t > s_0 + \frac{a_k}{\rho} \log \max\{1, h_0/H\}$, with $a_k = k + 1$ one-sided and $2k + 1$ two-sided. Following this iteration, each subsequent scan costs $O(|F_V| + |F_W| + |\text{supp } \alpha| + |\text{supp } \beta|)$ instead of $O(n + m)$.*

In words: since $r^2 \leq 2R(UB - LB)$, a point with margin τ_i is screened once the certified gap falls below $\tau_i^2/(18RL^2)$.

Proof of Theorem 11. Strong convexity gives $\|z - z^*\| \leq r$ for the screening radius $r = \sqrt{UB^2 - \max(LB, 0)^2}$ (τ is defined only if some point has positive margin; otherwise there is nothing to screen). Let $m^* = \min_j \langle z^*, v_j \rangle$, choose $j' \in F_V$ and set $v' = v_{j'}$; since $v_{\min}$ minimises $\langle z, \cdot \rangle$, $\langle z^*, v_{\min} \rangle - m^* = \langle z, v_{\min} - v' \rangle + \langle z^* - z, v_{\min} - v' \rangle \leq rL$. Hence $\langle z, v_i - v_{\min} \rangle = \langle z^*, v_i - v_{\min} \rangle + \langle z - z^*, v_i - v_{\min} \rangle \geq \tau_i - rL - r\|v_i - v_{\min}\|$, and the rule fires when this exceeds $r\|v_i - v_{\min}\|$, for which $\tau_i > 3rL$ suffices since $\|v_i - v_{\min}\| \leq L$. For the activation time, $r^2 \leq 2R(UB - LB)$ (for $LB \geq 0$, $r^2 = (UB - LB)(UB + LB)$ and $UB + LB \leq 2R$; for $LB < 0$, $r^2 = UB^2$ and $UB \leq 2(UB - LB)$), and $2R(UB - LB) \leq 2Rg^{\text{FW}}/UB \leq (2R/\delta^*)(2h + R\sqrt{2h}) \leq (4R^2/\delta^*)\sqrt{2h}$ (Lemma 10, with equality in the first step for the freshly evaluated bound; the last step uses $h \leq R^2/2$, which holds for every feasible iterate since $h \leq \frac{1}{2}\|z\|^2$); requiring $r^2 \leq \tau^2/(9L^2)$ yields $h_t < H$, and Theorem 6 with $\log(1/(1 - \rho)) \geq \rho$ converts it into the iteration bound (the $\max\{1, \cdot\}$ covers $h_0 \leq H$, where the condition holds from the start by monotonicity). The removal is executed at the first screening round after the condition holds; permanence follows from the safety of the rule. $\square$

Lemmas 3, 10 and 7, Proposition 9 and Theorem 11 are verified numerically on random and adversarial ensembles at machine precision (Section 6.2).

5 Exact reductions of the SVM variants

L_2 soft margin. The problem $\min \frac{1}{2}\|w\|^2 + \frac{C}{2} \sum_i \xi_i^2$ is the hard-margin problem on the augmented points $\tilde{x}_i = [x_i; e_i/\sqrt{C}]$, e_i the i -th standard basis vector, whose Gram matrix is $K + I/C$ (K the Gram matrix of the original points) and whose augmented hulls never intersect; let δ_C denote their distance. The augmentation is never materialised: the extra coordinates of an iterate are exactly its convex weights, so every augmented inner product is a cached original-space product plus a sparse diagonal correction, $\langle \tilde{p}, \tilde{v}_i \rangle = \langle p, v_i \rangle + \alpha_i/C$ and $\|\tilde{p}\|^2 = \|p\|^2 + \|\alpha\|^2/C$. The separator is recovered as $w = 2(p - q)/\delta_C^2$, $b = (\|\tilde{q}\|^2 - \|\tilde{p}\|^2)/\delta_C^2$, and the primal value is $P = 2/\delta_C^2$. Validation: the solver agrees with the SLSQP reference on the explicitly augmented problem to 10^{-10} across random instances and values of C ; the regularisation path δ_C is monotone in C ; on a9a ($C=0.1$) its certified primal 687.07 equals LIBLINEAR’s to five digits.

ν -SVM. The L_1 (hinge) soft margin is the nearest-point problem between *reduced* convex hulls $R(V, \mu) = \{\sum a_i v_i : \sum a_i = 1, 0 \leq a_i \leq \mu\}$, $\mu = 2/(\nu\ell)$ with $\ell = n + m$ the number of training points (Bennett and Bredensteiner, 2000). Two changes: the extreme-point oracle becomes a capped top- k blend (weight μ on the $\lfloor 1/\mu \rfloor$ best points in the search direction, via a partial sort, still $O(n)$), and the primary step is the capped MDM transfer with blended toward steps as fallback; caching and the reduced support-function lower bound carry over unchanged. Validation: agreement with the exact box-constrained QP to 10^{-9} – 10^{-14} and with LIBSVM’s NuSVC to $\sim 10^{-10}$ through the mapping $\delta^* = \|w_{\text{svc}}\|/\varsigma$, w_{svc} LIBSVM’s weight vector and ς the per-class sum of the rescaled dual coefficients. The reduced hull is a polytope, so (1) still applies, but its vertices are blends and the capped transfer is not the pairwise step analysed in §4; the analysis of this variant is left open.

Kernels. The algorithm touches data only through inner products, so kernelisation changes plumbing, not algorithm: Gram columns become kernel columns $\kappa(V, x_i)$, squared norms become kernel diagonals, iterates exist purely as convex weights (the exact cache refresh is reconstructed from the support’s cached kernel columns), and the classifier is the kernel expansion of the witness-pair bisector. A ridge $K + I/C$ gives the kernel L_2 margin by the same diagonal trick. Validation: the linear kernel reproduces the Euclidean solver to 10^{-12} ; RBF distances match the exact kernel QP to 10^{-9} and LIBSVM’s hard-margin solution, via $\|w_{\mathcal{H}}\|^2 = (\alpha \circ y)^\top K (\alpha \circ y)$, $w_{\mathcal{H}}$ the feature-space normal, for LIBSVM’s dual coefficients α and labels y , to 10^{-5} ; kernel SMO matches the QP to 10^{-8} .

6 Experiments

Setup. All solvers are validated against a reference QP solution (SLSQP from SciPy on the weight variables, function tolerance 10^{-14} , on instances small enough for it; distances agree to 10^{-9} or better). Stopping rule $\varepsilon = 10^{-3}$ on the certified relative gap unless stated; baselines use matched tolerances on the same objective, LIBLINEAR up to its intercept penalty (Section 6.1). Synthetic instances follow (Gupta and Kalantari, 2026) (two Gaussian clouds, $n = 5,000$ per class). Real data are the LIBSVM sets a9a, w8a, ijcnn1, gisette and covtype (100k subsample), standardised, official test sets where they exist, five seeds, 600s per solver and cell, all on one dedicated 32-vCPU Xeon 2.6 GHz machine (Section 6.1). “Our SMO” is our own implementation and K-ETA the kernelised ETA.

Blocks and screening (Section 6.1). Two effects separate in Table 6 (isolated runs). The guard and away fallback alone (guarded $k=1$) cut plain pairwise FW’s iterations 1.4 – $2\times$ but not its time (the two tie within CI); aggregation cuts iterations a further 4.2 – $4.4\times$, which buys wall-clock only where per-iteration $O(n)$ work, not column computation, dominates: $2.0\times$ over

Table 1: Regime battery on real data, five seeds, 600 s per solver, mean $\pm$ 95% CI. Kernel cells (RBF with width parameter $1/d$) use a class-balanced subsample of n_k points. Same objective per row: LIN, linear hard margin (ETA with screening vs LIBSVM’s $C=10^6$ surrogate; our SMO in brackets); KHM, feature-space hard margin (K-ETA vs the same surrogate); KL2, kernel L_2 margin, $C=1$ (K-ETA vs kernel SMO, exact objective; in brackets LIBSVM on the precomputed $K + I/C$ kernel, fit time + kernel construction). conv = seeds in which ETA certified; oracle = $O(nd)$ column (ETA) or row (SMO) evaluations, thousands. Run with the released code (fallback in every path, schedule of Lemma 7).

cell	dataset	n_k / d	ETA (s)	baseline (s)	conv	dist. ETA / base	acc. ETA / base	oracle
LIN	gisette	6000 / 5000	31.6 ± 2.8	36.5 ± 5.9 [259 $\pm$ 31]	5/5	3.6638 / 3.6640	.981 / .981	1.5 [11.1]
KHM	gisette	6000 / 5000	57.8 ± 8.7	68.1 ± 13.4	5/5	.05215 / .05215	.979 / .979	2.6
KHM	ijcnn1	9853 / 22	351 ± 109	1.1 ± 0.2	5/5	.00430 / .00430	.946 / .946	2.6
KHM	a9a	10000 / 123	> 600	19.8 ± 1.3	0/5	$\leq .00006$ / .0033	.707 / .752	5.9
KHM	covtype	10000 / 54	> 600	134 ± 3 (maxiter)	0/5	$\leq .00055$ / .00025	.694 / .712	7.5
KHM	w8a	6479 / 300	11.7 ± 32.4	—	5/5	hulls intersect	—	0.5
KL2	gisette	6000 / 5000	84.8 ± 15.4	243 ± 15 [$0.5 \pm 0.1 + 3$]	5/5	.06882 / .06882	.971 / .971	3.7 / 15.2
KL2	w8a	6479 / 300	8.7 ± 1.6	28.5 ± 1.3 [$0.6 \pm 0.1 + 1$]	5/5	.05682 / .05682	.975 / .975	3.6 / 20.2
KL2	a9a	10000 / 123	15.3 ± 2.8	27.0 ± 4.7 [$2.2 \pm 0.2 + 2$]	5/5	.03019 / .03019	.795 / .795	7.6 / 43.6
KL2	covtype	10000 / 54	21.4 ± 1.8	15.4 ± 0.6 [$2.1 \pm 0.1 + 2$]	5/5	.02609 / .02609	.792 / .792	9.1 / 56.2
KL2	ijcnn1	9853 / 22	7.7 ± 1.9	5.3 ± 0.4 [$1.1 \pm 0.1 + 2$]	5/5	.04790 / .04790	.959 / .959	4.6 / 26.1

guarded $k=1$ and MDM on ijcnn1 ($d=22$; $4.0\times$ over BPCG; $2.2\times$ and $4.5\times$ with the adaptive size of Section 7.4) and nothing on gisette ($d=5,000$), where columns are 87–93% of the time and all five configurations tie within CI at 31–38 s. $k=16$ is untuned (ρ is k -independent, an oversized block costs guard work, not progress; $k=4$ and 16 agree within CI). Screening gives 1.4 – $2.4\times$ at $\varepsilon = 10^{-5}$ and is neutral at 10^{-3} (Theorem 11 is a sufficient condition, which these runs stop before meeting).

Real data, sparse-support regime (Table 1). TA I first certifies separability (gisette yes, the other four no, in 0.1–35 s), a diagnostic no baseline produces. On gisette ($d=5,000$; 1,237 of 6,000 points in the support) ETA ties LIBSVM on the linear hard margin, both 7–8 $\times$ ahead of our SMO; K-ETA is 1.2 $\times$ faster than LIBSVM on the RBF hard margin and 2.9 $\times$ faster than kernel SMO on the kernel L_2 margin, with identical solutions. Across all five KL2 cells K-ETA matches kernel SMO’s solution with 4–6 $\times$ fewer kernel-column evaluations; in wall-clock it is 3 $\times$ faster where columns are expensive (gisette, w8a) and 1.4 $\times$ slower where they are cheap (ijcnn1, covtype). Against BPCG (Tsuji et al., 2022) with the same column cache and certificate (Table 7), block ETA needs 12–35 $\times$ fewer iterations, is 2.5–5.6 $\times$ faster on the five KL2 cells ($4.0\times$ on ijcnn1 in isolation) and ties on the gisette L_2 margin, where columns dominate. LIBSVM on the precomputed $K + I/C$ kernel (brackets in Table 1) reaches the same distances 2.5–24 $\times$ faster than K-ETA, kernel construction included: the KL2 wins are over column-oracle solvers only.

The KHM column shows where the geometry turns against the method. On ijcnn1 both return the same solution (819 support vectors), but K-ETA needs 1.6 – 2.7×10^5 iterations and 351 s against LIBSVM’s 1.1 s: the hulls are 0.004 apart in a space of diameter $\approx \sqrt{2}$, so a relative certificate needs an absolute gap of order 10^{-6} (a log-factor cost; δ/M is translation-invariant and unknown here), which suits LIBSVM’s second-order selection far better. On a9a and covtype the hulls are within 10^{-4} and 5×10^{-4} of touching: K-ETA times out with an uncertified iterate and LIBSVM’s $C=10^6$ surrogate stops at its KKT tolerance (iteration cap on covtype), so neither certifies anything. On w8a the hulls intersect (duplicate points, opposite labels); K-ETA certifies this in 12 s.

Real data, L_2 soft margins (Table 2). The same boundary appears within one problem class. Gisette is separable, so its L_2 margin has a sparse support and ETA converges to the certified gap in 30–45 s at every C , $7\times$ faster than our SMO. LIBLINEAR at matched tolerance (10^{-6}) takes 8 min to 1.5 h; at its default 10^{-4} it takes 5–66 s but stops 5–24% above the certified optimum with lower test accuracy (Table 4). ETA has no wall-clock advantage over LIBLINEAR on gisette; it adds the certificate that the fast answer is not the optimum. On the four overlapping datasets both geometric solvers hit the cap on every cell while LIBLINEAR finishes in 0.3–39 s (Table 5); at $C \geq 1$ the capped iterate is not usable, on covtype at no C . The uncapped traces (Figure 1) show the linear-rate signature of Theorem 6 with 60–65% of points in the support. Heuristically (δ has no known bound in terms of the class hulls), $n_s \approx 0.6n$ active atoms forming a near-regular simplex have pyramidal width $2/\sqrt{n_s}$ times the side (Lacoste-Julien and Jaggi, 2015), so $\rho \sim 1/n_s$ and the cost is $O(n^2)$ -type.

6.1 Experimental details

Solver settings and tolerance equivalence. ETA: block transfers with $k = 16$, guarded, away fallback, pairwise anti-zig-zag, stop at certified relative gap 10^{-3} , iteration cap 2×10^6 , 600 s cap. Tables 1, 6 and 7 were produced by the released code: the schedule of Lemma 7 and the case-(b) fallback in every path. Tables 3 and 8 and Figure 3 were rerun with it. Table 2 and the traces of Table 5 come from the submitted version, which used the unconditional V -then- W order (Table 3 shows the two schedules reach the same solutions with different iteration counts) and whose single-surviving-pair path took the possibly capacity-clipped MDM transfer directly, bypassing the fallback; the fix changes a block trajectory only on iterations with a single surviving pair (the $k \geq 4$ rows of Tables 6 and 8 have identical iteration counts under both versions), and every solution of Figure 3 agrees to 10^{-4} under both. Block assembly copies the k' cached columns into a $k' \times n$ array, an $O(k'n)$ copy, before the $O(k'^2)$ products; screening (every 25 iterations) only in the LIN cell, where it never activated at 10^{-3} (Theorem 11 gives a sufficient condition only, which these runs stop before meeting). Our SMO: maximal-violating-pair selection, KKT gap 10^{-3} , same caps; hard margin as $C = 10^{12}$; kernel SMO on $K + I/C$. LIBSVM on the precomputed $K + I/C$ kernel (scikit-learn `SVC`, `kernel='precomputed'`, $C=10^6$, tolerance 10^{-3} , five seeds; `src/libsvm_kl2.py`) solves the same kernel L_2 margin: the RBF kernel with $1/C$ on the diagonal is built with BLAS in 1–3 s and the fit takes 0.5–2.2 s, with distances equal to K-ETA’s to five digits. LIBLINEAR (scikit-learn 1.8 `LinearSVC`): squared hinge with parameter $C/2$, so that its objective equals $\frac{1}{2}\|w\|^2 + \frac{C}{2} \sum \xi_i^2$; tolerance 10^{-6} , chosen so that its final primal matches ETA’s certified value on every cell where ETA converges except gisette at $C=10$ (a9a 687.07 vs 687.07; gisette 0.14900 vs 0.14893, inside ETA’s certified 8×10^{-4} ; ijcn1 550.6 vs 550.5 at ETA gap 0.01); this tolerance is also why LIBLINEAR needs 18,000 iterations and over an hour on gisette at $C=1$, and 35,000 iterations and 1.5 h at $C=10$, where it nevertheless stops 5% above the certified optimum (0.1566 vs 0.1490). The gisette LIBLINEAR times in Table 2 are therefore the cost of matching ETA’s primal; Table 4 reports the same configuration at scikit-learn’s default tolerance 10^{-4} (five seeds, run separately without concurrent load): 5–66 s, but primals 5–24% above the certified optimum and lower test accuracy. Intercept scaling 100, so LIBLINEAR regularises $b/100$ where the geometric formulation leaves b free (the primals agree to five digits); iteration cap 2×10^5 . LIBSVM (scikit-learn `SVC`): tolerance 10^{-3} , 1000 MB cache, iteration cap 2×10^6 (reported as *maxiter* when hit); hard margins as $C = 10^6$; RBF kernel $\exp(-\|x - x'\|^2/d)$. Oracle counts: ETA counts Gram or kernel columns evaluated against a full class ($O(nd)$ or $O(n_k d)$ each); SMO counts kernel rows, the same unit. Columns are cached without eviction and the block Gram matrix Γ is assembled from four cached entries per pair, so peak cache memory is the column count times n_k doubles: 0.73 GB for the largest cell (covtype KL2, 9,100 columns), 0.61 GB for a9a KL2 and 0.36 GB for ijcn1 KL2; measured peak resident memory in the isolated runs of Table 6 is 0.74 GB on gisette (data-dominated) and 0.49–0.50 GB on ijcn1 KL2 (0.68 GB for BPCG).

Data and seeds. a9a, w8a, ijcn1 and gisette use the LIBSVM training and test files; covtype (binary, scaled) is subsampled to 100,000 points and split 75/25 by seed. Features are standardised on the training set. Kernel cells use a class-balanced subsample of at most 5,000 points per class (w8a’s minority class has 1,479 points, hence $n_k = 6,479$; gisette uses all 6,000). Seeds vary the covtype split, the kernel subsamples and the timing environment; on datasets with fixed splits the deterministic solvers return identical solutions across seeds, which is why several accuracy CIs are zero. Confidence intervals are t -based 95% intervals over seeds.

ETA on real data (L_2 soft margin, $C = 0.1$): certified gap to the 10^{-3} stopping rule

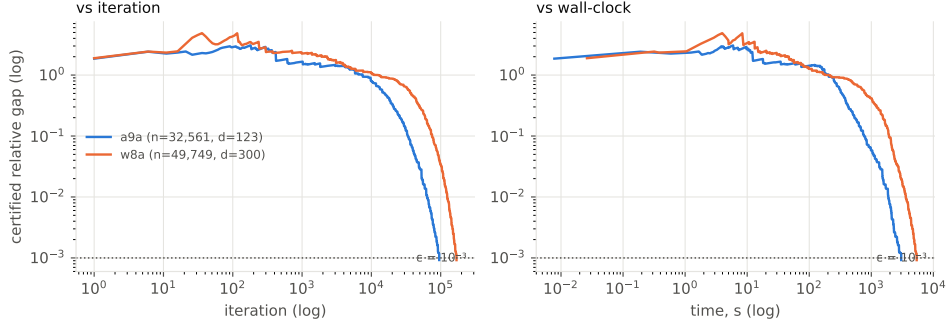


Figure 1: Certified relative gap of ETA on a9a and w8a (L_2 , $C = 0.1$, uncapped), versus iteration and wall-clock: linear-rate convergence in the dense-support regime, at $O(n^2)$ -type cost.

Hardware and concurrency. One dedicated 32-vCPU Intel Xeon 2.6 GHz Lightning Studio, 122 GB RAM, Linux, Python 3.12, NumPy 2.4 with OpenBLAS 0.3.31, `OMP_NUM_THREADS=1`. The L_2 battery ran 14 cells concurrently (its gisette, covtype and w8a $C=10$ cells were completed in later passes with 11–14 workers and no other load); the regime battery of Table 1 ran 12 cells concurrently with the ablation of Table 6, the BPCG runs of Table 7 and the LIBSVM kernel runs, 19 processes in all (the ablation was then rerun in isolation, below); Table 4 was made with no concurrent load; the five trials of Figure 3 ran as seven concurrent single-threaded jobs on an otherwise idle machine and reproduce the submitted run within its intervals; the absolute times of Tables 3 and 8 are about $3\times$ those of an earlier run with identical iteration counts, so their load is uncertain and only their within-table ratios should be read. The uncapped a9a/w8a runs in Figure 1 were made earlier on the same machine under a different load (LIBLINEAR took 12 and 48 s there against 4 and 23 s in the capped battery), so ratios across the two runs carry a factor of roughly two of uncertainty. Ratios within a battery share the load but not necessarily its effect, since dense BLAS, array scans and Python set operations contend for different resources, and the intervals measure the observed variation, not this systematic part; so only ratios whose intervals separate clearly should be read as results. The isolated rerun of Table 6 (`src/step_profile.py`, one configuration at a time on an otherwise idle machine) measures the effect: under the 19-process load the gisette MDM, guarded $k=1$ and $k=16$ configurations took 36.6, 27.7 and 37.8 s and ijcn1’s 16.0, 12.8 and 7.9 s; in isolation they take 30.1, 30.3, 30.8 and 15.3, 13.5, 7.0 s, and after the overhead fixes of Section 7.3 32.2, 30.7, 38.2 and 11.4, 11.2, 5.7 s (Table 6). The load thus manufactured one ratio ($1.3\times$ for guarded $k=1$ over MDM on gisette, a tie in isolation) while the ijcn1 ratios and the BPCG comparison (20.2 vs 5.9 s loaded, 21.1 vs 7.0 and 22.6 vs 5.7 s isolated) survive; the two isolated gisette runs also show that the machine’s column time itself moves by $\pm 15\%$ between runs.

Reference solver in the KL2 cells. The hinge-loss `SVC(rbf, C=1)`, a different objective, took 8.6 ± 2.9 (a9a), 3.4 ± 0.3 (covtype), 83.5 ± 13.1 (gisette), 1.1 ± 0.0 (ijcn1) and 4.9 ± 1.2 s

Table 2: L_2 soft-margin battery: five seeds, 600s per solver for ETA and our SMO (our own implementation; LIBSVM does not solve this objective), LIBLINEAR at tolerance 10^{-6} (Section 6.1) and unbounded; mean $\pm$ 95% CI, which on the fixed-split datasets (all but covtype) measures timing noise only. “cap” = timed out in every seed; ETA’s certified relative gap is reported at its stopping point (mean [min, max]); accuracies are those of the returned iterate.

dataset (n / d)	C	ETA (s)	SMO (s)	LIBLINEAR (s)	ETA gap at stop	acc. ETA / SMO / LIBLINEAR
gisette (6,000 / 5,000)	0.1	35.3 ± 6.8	243 ± 53	500 ± 41	8×10^{-4}	.981 / .981 / .981
	1	43.3 ± 5.3	295 ± 6	3916 ± 115	8×10^{-4}	.981 / .981 / .981
	10	33.2 ± 4.6	236 ± 36	5396 ± 197	10×10^{-4}	.981 / .981 / .977
a9a (32,561 / 123)	0.1	cap	cap	3.7 ± 0.8	0.15 [0.10, 0.23]	.849 / .845 / .849
	1	cap	cap	6.8 ± 1.7	1.26 [1.25, 1.28]	.810 / .780 / .849
	10	cap	cap	2.3 ± 0.5	1.83 [1.80, 1.86]	.635 / .763 / .849
ijcnn1 (49,990 / 22)	0.1	cap	478 ± 181	0.3 ± 0.1	0.010 [0.005, 0.016]	.918 / .918 / .918
	1	cap	cap	0.6 ± 0.2	0.68 [0.61, 0.73]	.901 / .917 / .918
	10	cap	cap	0.4 ± 0.1	1.34 [1.27, 1.40]	.656 / .852 / .918
w8a (49,749 / 300)	0.1	cap	cap	22.7 ± 4.3	0.57 [0.50, 0.66]	.987 / .985 / .987
	1	cap	cap	20.6 ± 6.6	1.17 [1.14, 1.21]	.923 / .979 / .987
	10	cap	cap	13.2 ± 1.1	2.02 [1.94, 2.14]	.720 / .971 / .987
covtype (100,000 / 54)	0.1	cap	cap	19.4 ± 15.7	1.04 [1.02, 1.05]	.568 / .660 / .756
	1	cap	cap	6.1 ± 3.6	1.42 [1.23, 1.73]	.508 / .541 / .756
	10	cap	cap	5.0 ± 1.8	2.01 [1.20, 2.37]	.517 / .514 / .756

(w8a), with test accuracies 0.789, 0.783, 0.977, 0.950 and 0.975, within one point of the L_2 solutions and on either side of them.

Step ablation, blocks and shrinking (synthetic). Iteration counts at $\varepsilon = 10^{-6}$ (single instances): $d=30$, $n=200$: toward 158,941, midpoint 148,996, away 106, pairwise 181; $d=100$, $n=1,000$: toward and midpoint fail within 300,000, away 15,886, pairwise 3,921. At the working tolerance 10^{-3} the pairwise/away gain over the midpoint heuristic is only $\sim 11\%$, which is why the heuristic sufficed in (Gupta and Kalantari, 2026). Block transfers cut iterations $3\text{--}25\times$ relative to single MDM (e.g. $43,486 \rightarrow 3,001$ at $k=16$ on a dense-support L_2 instance at $d=1,000$) and are the only configuration that reaches the certified tolerance on MNIST odd-vs-even. Figure 2 shows the screening speed-ups and Figure 3 the consolidated comparison, five trials per class (mean $\pm$ 95% CI): on the four hard-margin classes the optimised solver is $1.1\text{--}1.2\times$ faster than LIBSVM on average, with overlapping intervals (0.63 ± 0.04 vs 0.75 ± 0.15 s at $d=1,000$; 17.9 ± 1.6 vs 20.6 ± 3.7 s at $d=10,000$; 0.72 ± 0.13 vs 0.80 ± 0.18 s at $\varepsilon=10^{-5}$, where the paper-style configuration fails to converge in every trial; RBF 2.33 ± 0.06 vs 2.50 ± 0.12 s); on the synthetic L_2 margin ($d=1,000$, $C=1$) it ties LIBLINEAR (20.7 ± 9.7 vs 19.4 ± 4.4 s); on MNIST odd-vs-even it is $2.7\times$ slower (172 ± 20 vs 63 ± 22 s; single MDM never converges) and on the ν -SVM $1.7\times$ slower (61.0 ± 5.0 vs NuSVC’s 36.5 ± 5.6 s). Distances agree within tolerance on every row. An earlier run of the same script on a shared development container had shown the optimised solver $1.6\text{--}2.5\times$ ahead on six of the seven classes; that machine’s Python-to-C timing ratio did not reproduce on dedicated hardware, and only the dedicated numbers are reported here.

The guarded single pair already halves the iterations of plain MDM on ijcnn1 ($1.4\times$ on gisette): the drop steps it takes (on $0.2\text{--}2\%$ of its iterations) are what plain pairwise FW replaces by capacity-clipped swaps. Blocks then cut iterations a further $4.2\text{--}4.4\times$ ($k=4$, run only under load: 1,071 and 2,358 iterations, times within CI of $k=16$). Away fallbacks that drop an index occur on $0.2\text{--}36\%$ of guarded iterations, and their share grows with k : larger blocks add more indices per iteration, and the drops remove them. The time split says where fewer iterations buy time. On gisette, $87\text{--}93\%$ of every configuration’s time is Gram-column computation ($O(nd)$ per new receiver, $d=5,000$), so MDM, guarded $k=1$, block $k=16$, the adaptive size and BPCG all tie within CI at $31\text{--}38$ s. On ijcnn1 the columns cost about 1 s and the rest is per-iteration $O(n)$ work: the score update and line search inside the step, the certificate in the remainder, and least

Table 3: Two-sided schedules on the synthetic instances of Section 6.1 ($n=5,000$ per class, $k=16$): the unconditional V -then- W order used for the batteries versus the schedule of Lemma 7 (larger-gap side first, second side skipped after a drop). Drop steps and skipped second-side steps are counted over the run.

instance	schedule	time (s)	iterations	drops	skips	distance
hard margin, $d=1000$, $\varepsilon=10^{-5}$	unconditional	0.8	16	0	0	55.339365
hard margin, $d=1000$, $\varepsilon=10^{-5}$	Lemma 7	0.8	16	0	0	55.339365
L_2 overlap, $d=100$, $C=1$	unconditional	25.2	12,311	785	0	0.078400
L_2 overlap, $d=100$, $C=1$	Lemma 7	23.3	10,701	599	316	0.078400
L_2 overlap, $d=1000$, $C=1$	unconditional	12.9	2,901	182	0	0.578143
L_2 overlap, $d=1000$, $C=1$	Lemma 7	13.4	3,001	180	121	0.578143

Table 4: LIBLINEAR on gisette at its default tolerance (10^{-4}) versus the matched tolerance (10^{-6}) of Table 2; five seeds. ETA’s certified primal is 0.14835 / 0.14893 / 0.14899 at $C = 0.1$ / 1 / 10, reached in 30–45 s with test accuracy 0.981.

C	tol	time (s)	iterations	primal (vs certified)	accuracy
0.1	10^{-4}	66 ± 10	363	0.15648 (+5.5%)	.978
0.1	10^{-6}	500 ± 41	2,434	0.14835 (+0.0%)	.981
1	10^{-4}	5.5 ± 0.8	10	0.18427 (+24%)	.974
1	10^{-6}	3916 ± 115	18,063	0.14900 (+0.05%)	.981
10	10^{-4}	4.8 ± 0.1	8	0.18487 (+24%)	.973
10	10^{-6}	5396 ± 197	35,411	0.15663 (+5.1%)	.977

of all the scan itself. That work falls with the iteration count, though not in proportion, because a block step updates the scores with up to k' columns; the net effect is $2.0\times$ over MDM and $4.0\times$ over BPCG at $k=16$, and $2.2\times$ and $4.5\times$ with the adaptive size, which settles at $k=7-8$, the k^* of Section 7.3.

BPCG here is the blended pairwise step of (Tsuji et al., 2022) on the atoms $v_i - w_j$ of Z with the separable LMO of Lemma 1, exact line searches and the certificate of Lemma 10; its scores are maintained from the same cached columns, so column counts are comparable. It needs $12-35\times$ more iterations than block ETA on every cell, each an $O(n+m)$ scan; the block’s fewer scans buy $2.5-5.6\times$ in wall-clock where scans dominate (KL2) and nothing where columns do (gisette L_2).

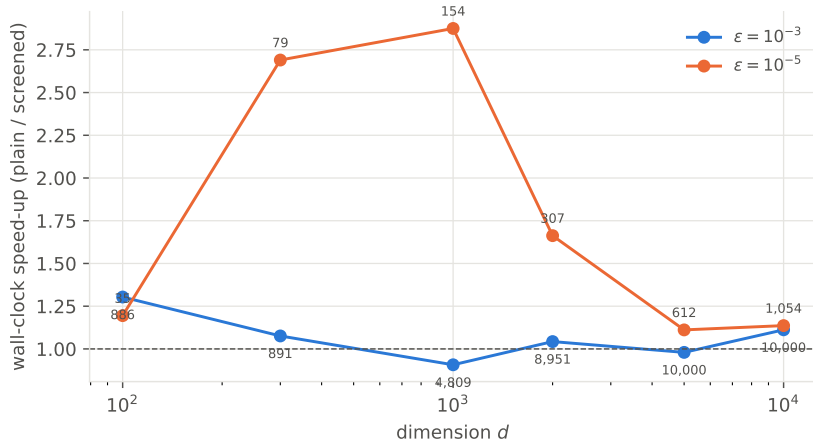


Figure 2: Safe screening: wall-clock speed-up vs dimension at two tolerances on the separated synthetic protocol; labels are surviving points of 10,000.

Table 5: Time to 1% of the optimum, seed-0 traces. The ETA columns measure *dual* progress and LIBLINEAR’s column *primal* progress, so this is not a matched primal comparison (the primal of ETA’s returned (w, b) is not recorded in the traces). For ETA the certified dual value $2/\text{UB}_t^2$ increases to the optimum, so $s_t = 1 - (2/\text{UB}_t^2)/P^*$ with P^* the best known primal (the lower of LIBLINEAR’s 10^{-6} value and ETA’s certified value); “certified 1%” is the first time the certificate itself proves $1 - \text{LB}_t^2/\text{UB}_t^2 \leq 1\%$. ETA’s a9a and w8a times are from the uncapped runs of Figure 1. For LIBLINEAR the entry is the time of the coarsest tolerance whose primal is within 1% of P^* ; where only 10^{-6} has been run it is an upper bound. On the remaining cells ETA is not within 1% at the 600s cap ($s = 30\text{--}99\%$).

dataset	C	P^*	ETA 1%	ETA 0.1%	cert. 1%	cert. 10^{-3}	LIBLINEAR 1%
gisette	0.1	0.14835	40 s	42 s	43 s	43 s	≤ 500 s (tol 10^{-6})
gisette	1	0.14893	44 s	46 s	47 s	48 s	≤ 3916 s (tol 10^{-6})
gisette	10	0.14899	34 s	36 s	37 s	37 s	none at 10^{-4} or 10^{-6}
a9a	0.1	687.07	811 s	1,413 s	2,121 s	3,030 s	4 s
w8a	0.1	108.80	2,084 s	2,960 s	4,162 s	5,386 s	23 s
ijcnn1	0.1	550.61	301 s	472 s	> 600 s	> 600 s ($s=0.05\%$)	0.3 s

6.2 Numerical verification of the analysis

Lemma 3 was checked on 4,000 random ensembles including heavily clipped and boundary cases (worst slack at machine precision); Proposition 9 on 500 near-orthogonal ensembles; Lemma 10 to 10^{-9} at arbitrary iterates; Lemma 7 on instrumented two-sided runs (no drop iteration added a support index, and every non-drop iteration met the gain bound with M replaced by the computable upper bound $\text{diam } V + \text{diam } W$); and Theorem 11 on instances whose optimal face was computed by the SLSQP reference: at every activating round all zero-weight non-face points were screened, and the final working set equalled the optimal face exactly. Over 5,000 instrumented live block steps none increased the objective. All checks are test files in the released code.

7 When does aggregation pay?

Fewer iterations are not the same as less work. This section separates the two: an exact expression for how much more progress a block makes than the single MDM step at the same iterate, a cost model for what the block costs, a theorem that bounds the drop steps by an identification phase rather than by the crude count of Theorem 6, and measurements of every quantity in the model on the five real cells of Section 6.

7.1 Progress per unit of work at one iterate

Consider a case-(a) iteration with k' retained pairs, all uncapped, so that $\gamma_j = \text{gap}_j/\|d_j\|^2$ and pair j alone would gain $\Delta_j = \text{gap}_j^2/(2\|d_j\|^2)$, with Δ_1 the MDM gain. Write

$$Q = \frac{\sum_{j \leq k'} \Delta_j}{\Delta_1}, \quad \kappa = \frac{D}{S}, \quad S = \sum_j \gamma_j \text{gap}_j = 2 \sum_j \Delta_j, \quad D = \left\| \sum_j \gamma_j d_j \right\|^2.$$

Q measures how much useful progress the block has beyond its leading pair and κ how much the directions lose (or gain) by being combined; orthogonal directions give $\kappa = 1$. Call the block η -diverse if $|\langle d_i, d_j \rangle| \leq \eta \|d_i\| \|d_j\|$ for $i \neq j$ and β -flat if $\Delta_j \geq \beta \Delta_1$ for every j .

Proposition 12 (Exact gain ratio). *At such an iteration the exact line search gives*

$$\frac{\Delta_B}{\Delta_1} = Q \cdot \begin{cases} 2 - \kappa, & \kappa \leq 1, \\ 1/\kappa, & \kappa \geq 1. \end{cases}$$

Table 6: Step ablation on real data with a time split, isolated runs (`src/step_profile.py`: three seeds, each configuration alone in its own process on an otherwise idle machine, run after the overhead fixes of Section 7.3): the gisette L_2 margin ($C=1$, $d=5,000$) and the ijcn1 kernel L_2 cell of Table 1 ($d=22$, $n_k=9,853$). MDM is plain pairwise FW (single transfers, no guard, no fallback); guarded $k=1$ is Algorithm 1 with a single pair (guard and case-(b) fallback active); auto is the adaptive block size of Section 7.4 (median k over the run given); BPCG as in Table 7. Split of the ETA time: scan = the score scan and pair selection; col = Gram or kernel column computation; step = the step itself (block assembly, guard, line search, cache and score update); rest = certificate and bounds. Drops = away fallbacks that removed a support index. Every run converged to the certified gap with identical distance and accuracy per dataset and seed. Peak resident memory: 0.74 GB for every gisette run and 0.49–0.50 GB (ETA) / 0.68 GB (BPCG) on ijcn1. The gisette column time varied by $\pm 15\%$ between two isolated runs on the same machine, which is the width of its intervals.

instance	step	k	time (s)	scan	col	step	rest	iterations	columns	drops
gisette L_2	MDM	1	32.2 ± 1.7	0.3	28.4	1.2	2.4	4,946	1,300	0
gisette L_2	guarded	1	30.7 ± 9.1	0.3	26.9	1.9	1.6	3,471	1,334	82
gisette L_2	block	16	38.2 ± 4.7	0.1	35.5	2.2	0.5	831	1,480	301
gisette L_2	auto	9	35.6 ± 3.1	0.1	33.1	1.9	0.5	986	1,451	194
gisette L_2	BPCG	—	33.8 ± 2.1					10,470	1,322	87
ijcn1 kernel L_2	MDM	1	11.4 ± 0.4	1.5	1.0	3.2	5.7	12,796	4,520	0
ijcn1 kernel L_2	guarded	1	11.2 ± 2.0	1.0	0.9	5.2	4.1	6,118	4,514	14
ijcn1 kernel L_2	block	16	5.7 ± 1.9	0.3	0.8	3.5	1.0	1,396	4,590	98
ijcn1 kernel L_2	auto	8	5.0 ± 0.5	0.3	0.8	2.8	1.1	1,734	4,570	75
ijcn1 kernel L_2	BPCG	—	22.6 ± 3.0					18,034	4,512	19

If the block is η -diverse then $\kappa \leq 1 + \eta(k' - 1)$; if it is β -flat then $Q \geq 1 + \beta(k' - 1)$; hence the guarded step gains at least

$$\max(\Delta_B, \Delta_1) \geq \Delta_1 \frac{1 + \beta(k' - 1)}{1 + \eta(k' - 1)}.$$

Proof. For uncapped pairs $\gamma_j \|d_j\|^2 = \text{gap}_j$, so $S = \sum_j \text{gap}_j^2 / \|d_j\|^2 = 2 \sum_j \Delta_j$ and $b_j^2 = \gamma_j^2 \|d_j\|^2 = \gamma_j \text{gap}_j$, whence $\sum_j b_j^2 = S$. The line search takes $\theta = \min(S/D, 1)$. If $D \geq S$ the optimum is interior and $\Delta_B = S^2/(2D) = (S/2)/\kappa = \sum_j \Delta_j/\kappa$; if $D < S$ then $\theta = 1$ and $\Delta_B = S - D/2 = (S/2)(2 - \kappa)$. Dividing by Δ_1 gives the two cases. For the bound on κ , $D = \sum_{i,j} \gamma_i \gamma_j \langle d_i, d_j \rangle \leq \sum_j b_j^2 + \eta \sum_{i \neq j} b_i b_j \leq (1 + \eta(k' - 1)) \sum_j b_j^2 = (1 + \eta(k' - 1))S$ by Cauchy–Schwarz on the cross terms; flatness gives $\sum_j \Delta_j \geq \Delta_1(1 + \beta(k' - 1))$. When $\kappa \leq 1$ the ratio is $Q(2 - \kappa) \geq Q$, and when $\kappa \geq 1$ it is Q/κ ; both are at least the stated bound. $\square$

Proposition 9 is the case $\beta = 1$ in disguise. The two numbers Q and κ are computed by the guard at every block step (from S , D and the per-pair gaps), so both are observable at no extra cost. Note the direction of the inequalities: the theorem’s η bounds the *maximum* absolute pairwise cosine of the retained directions, whereas the *effective correlation*

$$\hat{\eta} := \frac{\kappa - 1}{k' - 1}$$

that a block step reveals is a γ -weighted aggregate in which cross terms can cancel; the proof gives $\hat{\eta} \leq \eta$, and $\hat{\eta}$ can be negative. Likewise Q/k' close to 1 says the average pair is as good as the leading one, not that every pair is (β -flatness). The empirical model of Section 7.3 is therefore stated in Q and κ , the quantities the guard measures, and the theorem’s assumptions (all retained pairs uncapped, maximum cosine at most η , minimum relative gain at least β) are recorded separately.

Table 7: BPCG (Tsuji et al., 2022) versus block ETA ($k=16$) on the same objective with the same certificate ($\varepsilon=10^{-3}$), the same column cache and the same 600 s cap (`src/bpcg.py`); three seeds, run concurrently with Table 1; the isolated rerun of Table 6 gives `ijcnn1 KL2` 22.6 ± 3.0 vs 5.7 ± 1.9 s and `gisette L2` 33.8 ± 2.1 vs 38.2 ± 4.7 s. Every cell converged; distance and test accuracy agree per cell to five digits. Columns are kernel or Gram columns computed; drops count BPCG’s capacity-clipped local steps and ETA’s away drops.

cell	solver	time (s)	iterations	columns	drops
<code>gisette L2, C=1</code>	BPCG	34.2 ± 20.4	10,470	1,322	87
<code>gisette L2, C=1</code>	ETA	34.3 ± 15.0	831	1,480	301
<code>gisette KL2</code>	BPCG	213 ± 28	8,702	3,641	3
<code>gisette KL2</code>	ETA	83.9 ± 19.0	251	3,653	4
<code>w8a KL2</code>	BPCG	22.3 ± 5.8	12,342	3,577	12
<code>w8a KL2</code>	ETA	5.0 ± 2.2	603	3,591	50
<code>a9a KL2</code>	BPCG	65.4 ± 5.7	28,653	7,605	24
<code>a9a KL2</code>	ETA	11.7 ± 2.2	1,489	7,616	35
<code>covtype KL2</code>	BPCG	90.6 ± 11.2	38,422	9,065	15
<code>covtype KL2</code>	ETA	20.3 ± 1.1	3,336	9,074	26
<code>ijcnn1 KL2</code>	BPCG	20.2 ± 1.6	18,034	4,512	19
<code>ijcnn1 KL2</code>	ETA	5.9 ± 1.1	1,396	4,590	98

Table 8: Block-size ablation on the synthetic instances (current code): single MDM ($k=1$) versus blocks of k transfers. Iterations fall with k ; wall-clock does not fall in proportion, because each block iteration touches more columns.

instance	k	time (s)	iterations	columns	drops
hard margin, $d=1000$, $\varepsilon=10^{-5}$	1	0.6	301	104	0
hard margin, $d=1000$, $\varepsilon=10^{-5}$	4	0.6	36	103	0
hard margin, $d=1000$, $\varepsilon=10^{-5}$	16	0.5	16	103	0
hard margin, $d=1000$, $\varepsilon=10^{-5}$	32	0.6	16	109	0
L_2 overlap, $d=100$, $C=1$	1	26.3	72,546	1,345	0
L_2 overlap, $d=100$, $C=1$	4	27.2	24,606	1,632	416
L_2 overlap, $d=100$, $C=1$	16	23.9	10,701	2,404	599
L_2 overlap, $d=100$, $C=1$	32	27.4	7,606	3,076	305
L_2 overlap, $d=1000$, $C=1$	1	24.8	43,486	1,242	0
L_2 overlap, $d=1000$, $C=1$	4	17.7	10,071	1,330	177
L_2 overlap, $d=1000$, $C=1$	16	13.5	3,001	1,348	180
L_2 overlap, $d=1000$, $C=1$	32	13.1	1,741	1,396	170

Cost. Per iteration the solver pays a scan and certificate cost that does not depend on k , a step cost that grows with the number of retained pairs (the $k' \times n$ column copies of block assembly and the k' -column score update), and one Gram or kernel column for every receiver or donor not in the cache. Write

$$C_B = a + b k' + c_{\text{col}} m_B, \quad C_1 = a + b + c_{\text{col}} m_1,$$

for the block and the single step from the same state, where $m_B \geq m_1$ count the column misses of the two candidates ($m_1 \leq 1$: only the MDM receiver can be new) and a, b, c_{col} are timing coefficients. They are fitted, not assumed: they depend on n, d , the representation (dense, kernel) and the hardware, and Section 7.3 fits them per cell. The model is a timing approximation of the implementation, not an upper bound on computational work. The guarded block improves progress per unit of work at this iterate exactly when $\max(\Delta_B, \Delta_1)/\Delta_1 > C_B/C_1$; by Proposition 12 a sufficient condition is

$$\frac{1 + \beta(k' - 1)}{1 + \eta(k' - 1)} > \frac{a + b k' + c_{\text{col}} m_B}{a + b + c_{\text{col}} m_1}. \quad (2)$$

Two regimes are visible in (2). If c_{col} dominates a and b , the right-hand side is governed by the miss counts: the block loses when it touches more new points than the single step ($m_B > m_1$)

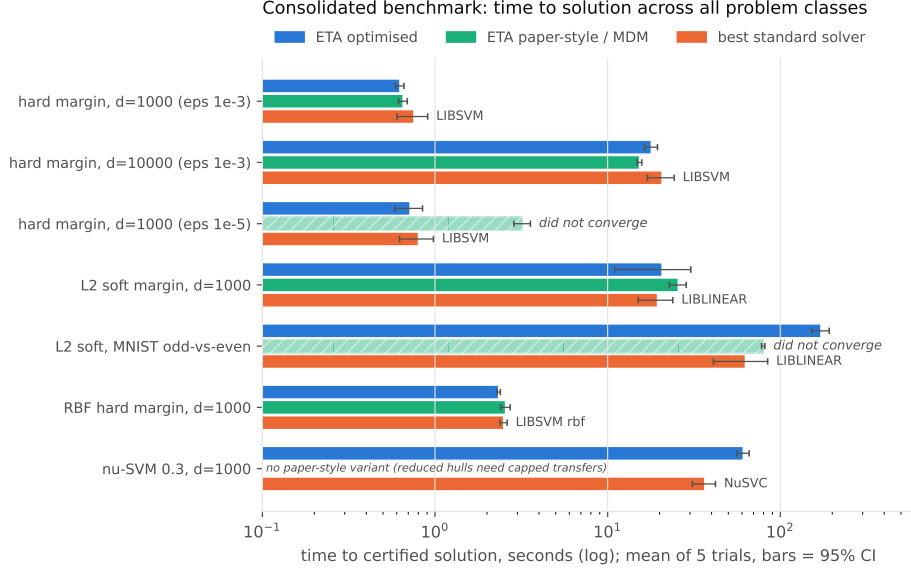


Figure 3: Time to certified solution on seven problem classes: optimised ETA vs the paper-style configuration vs the best standard solver per class; five trials per class, mean with 95% CI. The ν -SVM row has no paper-style bar because the toward-step algorithm of (Gupta and Kalantari, 2026) has no reduced-hull variant (reduced hulls need capped transfers).

and can still win by the full gain ratio when both need the same column ($m_B = m_1$), so column dominance limits the gain to the iterations on which the block’s extra receivers are cached; it does not exclude it. If the per-iteration terms dominate, the right-hand side is $(a + bk')/(a + b)$, linear in k' , while the left-hand side saturates at β/η ; the ratio of the two is maximised at

$$k^* = \sqrt{\frac{a(1-\eta)}{b\eta}} \quad (\beta = 1), \quad (3)$$

the block size at which the marginal cost of one more pair equals its marginal gain.

7.2 Drops are an identification cost

Theorem 6 charges up to k drop steps per non-drop iteration, which would cancel a k -fold gain in progress. The measurements below show that drops in fact happen while the support is being built and stop once it has settled. The following theorem proves this under the standard non-degeneracy assumptions of active-set identification (Bomze et al., 2020; Garber, 2020): after an explicit iteration, every step that removes a non-face point is a drop, no non-face point enters, and once the supports lie in the optimal faces and the weights are close to the optimal ones, no step is a drop at all.

Assumption 13. (A1) Strict complementarity: $F_V = \text{supp } \alpha^*$ and $F_W = \text{supp } \beta^*$, and $\alpha_{\min} > 0$ is the smallest optimal weight. (A2) Non-degeneracy: the witness pair (p^*, q^*) is unique, the vertices of each optimal face are affinely independent (so α^*, β^* are unique), and there is $\sigma > 0$ with $\|\alpha - \alpha^*\|_1 + \|\beta - \beta^*\|_1 \leq \|(p - q) - z^*\|/\sigma$ for all $p \in \text{conv}(F_V)$, $q \in \text{conv}(F_W)$ with weights α, β on the faces. Let $d_{\min} > 0$ be the smallest distance from a face vertex to the affine hull of the other vertices of its face.

Affine independence and uniqueness of the witness pair make the linear map $(\alpha - \alpha^*, \beta - \beta^*) \mapsto (p - q) - z^*$ injective on the subspace $\{\sum_i (\alpha_i - \alpha_i^*) = 0 = \sum_j (\beta_j - \beta_j^*)\}$; if $\sigma_2 > 0$ is the smallest

singular value of the matrix $[V_{F_V}, -W_{F_W}]$ restricted to that subspace, then $\sigma = \sigma_2/\sqrt{|F_V| + |F_W|}$ works, the square root being the conversion from the Euclidean to the ℓ_1 norm of the weights.

Recall $r_t = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2} \geq \|z_t - z^*\|$ and, from the proof of Theorem 11, $r_t^2 \leq (4R^2/\delta^*)\sqrt{2h_t}$, so that $r_t \leq \varrho$ whenever $h_t \leq \varrho^4\delta^{*2}/(32R^4)$. Let s_t be the total support size, $a_k = k + 1$ (one-sided) or $2k + 1$ (two-sided schedule), and define

$$H_{\text{id}} = \min \left\{ \frac{\tau^4\delta^{*2}}{41472R^4L^4}, \frac{\tau^2}{12L^2}, \frac{\tau}{12}, h_3 \right\}, \quad h_3 = \max\{h : 2h + R\sqrt{2h} \leq \tau/6\},$$

$$H_w = \min \left\{ H_{\text{id}}, \frac{(\sigma\alpha_{\min}/2)^4\delta^{*2}}{32R^4}, \frac{(\alpha_{\min}d_{\min}^2/(2L))^4\delta^{*2}}{32R^4} \right\}.$$

Theorem 14 (Identification and drop count). *Let $\delta^* > 0$, let τ, L, R be as in Theorem 11 and let the guarded block-transfer algorithm run with any $k \geq 1$ under either schedule of Theorem 6. Let t_{id} be the first iteration with $h_t < H_{\text{id}}$ and N the number of non-face support indices at t_{id} .*

- (i) *For every $t \geq t_{\text{id}}$: if either side's support contains a non-face point, the step taken is a drop of a non-face point; and no non-face point enters either support. Hence the supports lie in $F_V \times F_W$ from iteration $t_{\text{id}} + N$ on, and $N \leq s_{t_{\text{id}}} \leq s_0 + a_k t_{\text{id}}$.*
- (ii) *Under Assumption 13, let t_w be the first iteration with $h_t < H_w$. No iteration $t \geq \max(t_w, t_{\text{id}} + N)$ is a drop, and no block pair is capped at such an iteration. Consequently every drop occurs before $t_w + N$, the number of drops is at most $a_k(t_w + N) + s_0$, and $t_w \leq s_0 + 1 + \frac{a_k}{\rho} \log \max\{1, h_0/H_w\}$.*
- (iii) *For $t \geq \max(t_w, t_{\text{id}} + N)$ every iteration contracts: $h_{t+1} \leq (1 - \rho_t)h_t$ with $\rho_t \geq \rho$, and $\rho_t \geq \rho_k := \min\{1, 4\rho(1 + \beta(k' - 1))/(1 + \eta(k' - 1))\}$ when the iteration is case (a) with an η -diverse, β -flat block.*

Proof. Throughout, $s_i = \langle g, v_i \rangle = -\langle z, v_i \rangle$ on side V and $s_j = \langle z, w_j \rangle$ on side W ; we argue on side V , side W being symmetric with the maximum in place of the minimum.

Three facts. (F1) If $r \leq \tau/(6L)$ then for $i \notin F_V$ and $j \in F_V$, $s_j - s_i = \langle z, v_i - v_j \rangle = \langle z^*, v_i - v_j \rangle + \langle z - z^*, v_i - v_j \rangle \geq \tau_i - rL \geq 5\tau/6$, while for $i, j \in F_V$, $|s_i - s_j| \leq rL \leq \tau/6$; in particular the global argmax of s is a face point. (F2) Let ω_t be the total weight on non-face points of both sides. By convexity $h_t \geq \langle \nabla F(z^*), z_t - z^* \rangle = \langle z^*, z_t - z^* \rangle$; writing $\langle z^*, v_i \rangle = m_V + \tau_i$, $\langle z^*, w_j \rangle = M_W - \tau_j$ and using $\|z^*\|^2 = \langle z^*, p^* - q^* \rangle = m_V - M_W$ (the optimal weights live on the faces, where the margins vanish), $\langle z^*, z_t - z^* \rangle = \sum_i \alpha_i \tau_i + \sum_j \beta_j \tau_j \geq \tau \omega_t$. Hence $\omega_t \leq h_t/\tau$. (F3) By the radius bound quoted above, $h_t < H_{\text{id}}$ implies $r_t \leq \tau/(6L)$, $\omega_t \leq \tau/(12L^2)$, $\omega_t \leq 1/12$ and $2h_t + R\sqrt{2h_t} \leq \tau/6$.

(i) *Identification.* Let $h_t < H_{\text{id}}$ and suppose $\text{supp } \alpha \not\subset F_V$. The worst active donor u minimises s over the support, so by (F1) $u \notin F_V$. The away gap of side V is $g_V^A = \sum_i \alpha_i(s_i - s_u) \geq \sum_{i \in F_V} \alpha_i \cdot 5\tau/6 \geq (1 - \omega_t)5\tau/6 \geq 55\tau/72$. A side whose support lies in its face has $g^{\text{PW}} \leq g^A + g^{\text{FW}} \leq rL + (2h + R\sqrt{2h}) \leq \tau/3$ by (F1) and (F3), so the side with the larger pairwise gap, which the schedule steps first, is a side with a non-face support point whenever one exists. On that side pair 1 is (r^*, u) with $r^* \in F_V$ the global argmax, so $\text{gap}_1 \geq 5\tau/6$ and the uncapped transfer would be $\text{gap}_1/\|d_1\|^2 \geq 5\tau/(6L^2) > \tau/(12L^2) \geq \omega_t \geq \alpha_u$: pair 1 is capped and the iteration is case (b). The away step on u has derivative $\langle g, p - v_u \rangle = g_V^A \geq 55\tau/72$ and curvature $\|p - v_u\|^2 \leq L^2$, so its unconstrained length is at least $55\tau/(72L^2)$, while its maximal length is $\alpha_u/(1 - \alpha_u) \leq \frac{12}{11}\alpha_u \leq \tau/(11L^2)$; the away step is boundary-clipped, i.e. a drop of u , and it adds no index. If instead neither side has a non-face support point, all donors are face points and a non-face receiver r has gap $= s_r - s_u \leq -5\tau/6 < 0$ against any of them, so it is discarded (the discard rule of Algorithm 1); the toward candidate is $r^* \in F_V$. Under the two-sided schedule the second side steps only on non-drop iterations, when neither side has a non-face point, and the same argument applies. Thus for $t \geq t_{\text{id}}$ (monotonicity of h) the

set of non-face support indices never grows and loses one member at every iteration while it is nonempty, which proves (i); the bound on N is the support count of Theorem 6.

(ii) *No late drops.* Let $t \geq \max(t_w, t_{\text{id}} + N)$, so the supports lie in the faces and, by (F3) applied to H_w , $r_t \leq \min\{\tau/(6L), \sigma\alpha_{\min}/2, \alpha_{\min}d_{\min}^2/(2L)\}$. By (A2), $\|\alpha_t - \alpha^*\|_\infty \leq r_t/\sigma \leq \alpha_{\min}/2$, so every support weight is at least $\alpha_{\min}/2$ (A1). Any pair (r, u) of face points has gap $\leq rL$ by (F1) and $\|v_r - v_u\| \geq d_{\min}$, so its uncapped transfer is at most $rL/d_{\min}^2 \leq \alpha_{\min}/2 \leq \alpha_u$: no pair is capped, the iteration is case (a), no away step is taken, and a block transfer removes at most $\gamma_j < \alpha_{u_j}$ from each donor, so no index leaves the support. Drops at $t \geq t_w$ are therefore among the N identification drops of (i), which occur at the consecutive iterations $t_{\text{id}}, \dots, t_{\text{id}} + N - 1$; hence all drops occur before $t_w + N$, and the count of Theorem 6 over the first $t_w + N$ iterations gives the bound. The bound on t_w is Theorem 6 with $\log(1/(1 - \rho)) \geq \rho$.

(iii) *Rate.* A non-drop iteration gains at least the amount of Lemma 4, which with Lemma 5 is ρh_t . In case (a) the guard takes $\max(\Delta_B, \Delta_1)$; by Proposition 12 this is at least $\Delta_1(1 + \beta(k' - 1))/(1 + \eta(k' - 1))$, and in case (a) the proof of Lemma 4 gives $\Delta_1 \geq (g^{\text{PW}})^2/(8M^2) \geq 2\delta^2 h_t/(8M^2) = 4\rho h_t$ by Lemma 5. A gain cannot exceed h_t , whence the minimum with 1. $\square$

Theorem 14 says that drops are paid during identification and never afterwards: the worst case k drops per iteration of Theorem 6 applies only up to $t_w + N$, and the k -dependence of the drop count enters only through the support that is built up to then. This is exactly the shape of the measurements in Figure 4. The constants are those of Theorem 11 and of active-set identification and are pessimistic; the measured drop counts below are 0.3–70% of the final support size.

Corollary 15 (Conditional work bound). *Run the algorithm with requested block size k and let $k'_t \leq k$ be the number of pairs retained at iteration t (on a case-(b) or drop iteration set $k'_t = 1$). Let $t_1(k) := \max(t_w, t_{\text{id}} + N)$ as in Theorem 14, $N_{\text{cols}}(k)$ the number of distinct columns computed over the whole run, and suppose every case-(a) iteration $t \geq t_1(k)$ is η -diverse and β -flat. Then for every $T \geq t_1(k)$,*

$$h_T \leq h_{t_1(k)} \prod_{t=t_1(k)}^{T-1} (1 - \rho_t), \quad \rho_t \geq \rho_{k'_t} := \min \left\{ 1, 4\rho \frac{1 + \beta(k'_t - 1)}{1 + \eta(k'_t - 1)} \right\}$$

on case-(a) iterations and $\rho_t \geq \rho$ otherwise, and the timing model charges the run

$$W(k) \approx c_{\text{col}} N_{\text{cols}}(k) + \sum_{t < T} (a + bk'_t).$$

In particular, if $h_{t_1(k)} > \varepsilon$ and $k'_t \geq \underline{k}$ on every case-(a) iteration after $t_1(k)$, then $h_T \leq \varepsilon$ for $T = t_1(k) + \lceil \log(h_{t_1(k)}/\varepsilon)/\rho_{\underline{k}} \rceil$ when $\rho_{\underline{k}} < 1$ (and $T = t_1(k) + 1$ when $\rho_{\underline{k}} = 1$), and $W(k) \lesssim c_{\text{col}} N_{\text{cols}}(k) + (a + bk)[t_1(k) + \lceil \log(h_{t_1(k)}/\varepsilon)/\rho_{\underline{k}} \rceil]$; if $h_{t_1(k)} \leq \varepsilon$ the run is over at $t_1(k)$.

Proof. Theorem 14(iii) with k' read as k'_t gives the product; $\log(1/(1 - \rho)) \geq \rho$ turns it into the iteration count, and the cost model charges $a + bk'_t \leq a + bk$ per iteration. $\square$

The corollary is an absolute bound for one run. It does not by itself compare two block sizes: $N_{\text{cols}}(k)$, $t_1(k)$ and $h_{t_1(k)}$ all depend on the trajectory, and comparing the multiplicative factors of two upper bounds is not a proof that one algorithm is faster. What it does say is which factor aggregation can act on. The per-iteration term after identification carries the factor $(a + bk'_t)/\rho_{k'_t}$, and if $k'_t \equiv k$ and $\beta = 1$ this is proportional to $(1 + \eta(k - 1))(a + bk)/k$, minimised at k^* of (3); the column term is not touched by the step rule at all. Whether N_{cols} , t_1 and h_{t_1} are in fact insensitive to k is an empirical question, answered below for the five cells (column counts within 20% across k ; drops and misses concentrated in the same early deciles for every k). A comparative speed-up theorem would need those three quantities bounded uniformly in k , which we do not prove.

Table 9: The model of Corollary 15 fitted per cell (seed 0, isolated runs). $\hat{\eta}$ is the median over $k \in \{4, 16, 32\}$; the three estimates agree within ± 0.03 on every cell. k^* from (3). Times for $k = 1/4/16/32$; “column share” is the fraction of the $k=16$ run spent computing columns; the last column is the speed-up of the best k over the guarded single pair.

cell	$\hat{\eta}$	a (ms)	b (ms)	c_{col} (ms)	k^*	time (s), $k = 1/4/16/32$	column share	best k vs $k=1$
gisette L_2	0.07	1.3	0.080	21.30	14	33.0 / 29.2 / 33.9 / 38.0	93%	1.1 $\times$ (4)
ijcnn1 KL2	0.26	1.7	0.098	0.23	7	11.6 / 6.1 / 6.1 / 7.6	17%	1.9 $\times$ (4)
w8a KL2	0.17	1.3	0.077	0.88	9	9.7 / 5.3 / 4.4 / 4.6	64%	2.2 $\times$ (16)
a9a KL2	0.11	2.5	0.111	0.62	13	30.5 / 14.2 / 10.8 / 11.7	41%	2.8 $\times$ (16)
covtype KL2	0.29	2.8	0.102	0.35	8	38.9 / 21.0 / 18.2 / 21.7	17%	2.1 $\times$ (16)

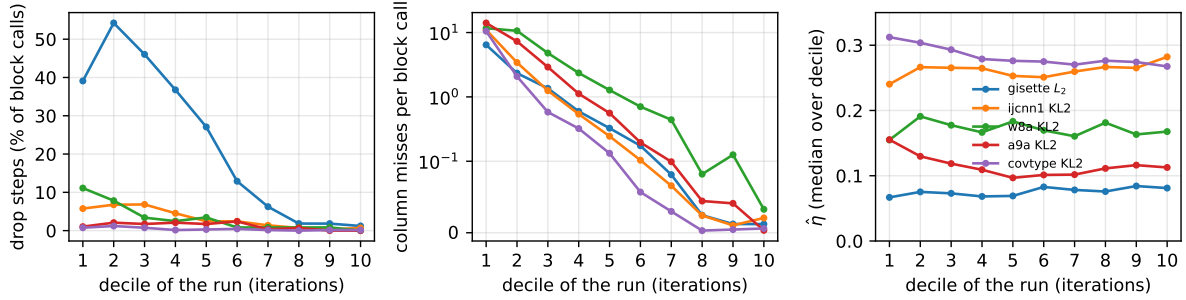


Figure 4: Along the run at $k = 16$ (deciles of the iteration count, one line per cell): fraction of block calls that are drops, column misses per block call, and the effective correlation $\hat{\eta}$. Drops and misses are paid while the support is built and become rare once it has settled, the shape Theorem 14 describes; $\hat{\eta}$ is constant along the run.

7.3 Measurements

Every quantity in Proposition 12, the cost model and Theorem 14 was recorded on the five real cells of Table 1 (gisette L_2 and the kernel L_2 cells of ijcnn1, w8a, a9a and covtype, seed 0; `src/block_diagnostics.py`, one process at a time on an otherwise idle machine, $k \in \{1, 4, 16, 32\}$). At every block step the solver logs Δ_B , Δ_1 , Q , κ , the candidate taken, the capped pairs, the column misses, the support size and a time split of the step next to a dry-run timing of the single MDM update from the same state. Table 9 fits the model per cell: $\hat{\eta} = (\kappa - 1)/(k' - 1)$ (median over uncapped block steps), a and b by least squares of the non-column time per iteration on k , c_{col} as column time per column.

Four facts follow.

- $\hat{\eta}$ is stable on each cell. The effective correlation does not move with k nor along the run (Figure 4, right), although it is a statistic of the iterates and the pairing rule rather than of the data alone; it is 0.07 on gisette, 0.11 on a9a, 0.17 on w8a, 0.26 on ijcnn1 and 0.29 on covtype, and 0.004 on the synthetic dense-support instance of Table 8. Q/k' is 0.85–0.98 throughout: the useful progress beyond the leading pair is never the limit, the loss to combination is. The measured gain ratios (3.4/6.9/7.5 on gisette, 2.1/2.8/3.0 on ijcnn1 for $k = 4/16/32$) follow $k/(1 + \hat{\eta}(k - 1))$ within 25% and saturate at about $1/\hat{\eta}$.
- k^* is 7–14 on every cell, because a/b is 16–27 throughout and $\hat{\eta} \in [0.07, 0.29]$: a fixed block size of 8–16 is within 16% of the best measured time on all five cells ($k=16$ against the best size: 16% on gisette, 0–5% elsewhere), which is why the untuned $k = 16$ of Section 6 never lost to another size by more than that.
- The wall-clock gain is bounded by the column share. The block cannot touch the column term of Corollary 15 (N_{cols} varies by under 20% across k on every cell), so its speed-up over the guarded single pair is 1.1 $\times$ where columns are 93% of the run (gisette), 2.2 $\times$ at

Table 10: The theorem’s assumptions against the measured mechanism at $k = 16$ (seed 0; `src/block_diagnostics.py` on a second machine, with iteration and drop counts identical to the Studio run). Over the block calls with every retained pair uncapped: median maximum absolute pairwise cosine of the retained directions (the η of Proposition 12), median minimum relative pair gain (its β), and the gain ratio the proposition then guarantees, $(1 + \beta(k' - 1))/(1 + \eta(k' - 1))$; against the effective correlation $\hat{\eta}$, the model ratio $k'/(1 + \hat{\eta}(k' - 1))$ and the measured median gain ratio $\max(\Delta_B, \Delta_1)/\Delta_1$.

cell	all pairs uncapped	max cos	β	guaranteed	$\hat{\eta}$	model	measured
gisette L_2	43%	0.33	0.53	1.5	0.08	7.4	6.9
ijcnn1 KL2	73%	0.42	0.80	1.8	0.26	3.2	2.8
w8a KL2	76%	0.33	0.77	2.1	0.17	4.4	3.6
a9a KL2	89%	0.28	0.70	2.2	0.12	5.8	4.6
covtype KL2	96%	0.42	0.82	1.8	0.28	3.0	2.7

64% (w8a), $2.8\times$ at 41% (a9a) and $1.9\text{--}2.1\times$ at 17% (ijcnn1, covtype), where the iteration gain has already saturated at $1/\hat{\eta}$. The model’s time predictions are within 10% of the measurements at $k \leq 4$ and on gisette; at $k = 16\text{--}32$ on the kernel cells they are 10–40% too high, because the runs need fewer iterations than the one-state ratio predicts: the trajectory effect is in the block’s favour, and the model is conservative.

- *Drops are front-loaded* (Figure 4, left). On gisette at $k = 16$ half of the block calls in the first three deciles are drops and 1–2% in the last three; the kernel cells show the same shape at lower levels; nearly every case-(b) call is a drop, so the best-of-four branch is rare. Column misses follow the same curve, so on column-bound cells the whole column bill is paid during identification whichever k is used. Total drops per final support point are 0.13/0.13/0.48/0.69 on gisette and 0.003–0.10 on the kernel cells for $k = 1/4/16/32$: proportional to the support, growing with k , and an order of magnitude more frequent on the sparse linear cell, where fresh receivers of small weight become the worst donor at once.

The theorem’s constants are not what the mechanism runs on. Table 10 records the assumptions of Proposition 12 on the real cells. The maximum pairwise cosine is 0.3–0.4 on every cell, so the η of the proposition does not distinguish the cells at all, whereas the effective correlation $\hat{\eta}$ spans 0.07–0.29 and does; the minimum relative gain is 0.5–0.8; and at $k = 16$ only 43–96% of the block calls have all pairs uncapped, so the proposition applies to a minority of the calls on gisette. The gain ratio the proposition guarantees from these constants is 1.1–2, against measured medians of 2.7–6.9; the exact identity Q/κ (the “model” column) reproduces the measurement because it uses the γ -weighted aggregate in which the cross terms largely cancel, not the worst pair. The right reading of Proposition 12 is therefore its first, exact, half; the η -diverse, β -flat bound is a worst case that real blocks beat by a factor of 2–4.

The measurements are from one seed per cell and the fits are retrospective (the coefficients are estimated from the same four runs they describe); the ratios they support (Tables 6 and 7) are reproduced across three seeds in isolated runs, and Section 7.4 is the prospective test. On the identification mechanism the figure is consistent with Theorem 14 but does not verify it: the theorem’s thresholds $H_{\text{id}}, H_{\text{w}}$ lie far below the stopping tolerance of these runs, and late drops, though rare, do occur. Section 7.5 tests the theorem where its constants can be computed.

7.4 An adaptive block size

Every quantity in (3) is observable while the solver runs: κ is computed by the guard at every block step, so $\hat{\eta} = (\kappa - 1)/(k' - 1)$ costs nothing, and a/b is a property of the implementation (15–30 in Table 9). The released code therefore offers `block_size='auto'`: every 50 iterations it sets k to $\sqrt{(a/b)(1 - \hat{\eta})/\hat{\eta}}$, clipped to $[1, k_{\text{max}}]$, with $\hat{\eta}$ the median over the window, and

Table 11: Adaptive block size on synthetic L_2 instances of varying directional diversity (one process, seed 0, a/b set to 20, $k_{\max} = 32$). Times for guarded $k=1$ and fixed $k = 4/16/32$; “auto” reports the size the rule settled on and its time.

d	$\hat{\eta}$	k^*	$k=1$ (s)	$k=4$	$k=16$	$k=32$	auto: k , time (s)
20	0.31	7	1.3	1.0	1.1	1.3	6, 1.1
50	0.10	13	2.1	1.2	1.0	1.3	12, 1.1
100	0.03	25	3.5	1.8	1.1	1.2	24, 1.1
300	0.00	> 32	1.7	0.7	0.4	0.4	32, 0.4
1,000	0.00	> 32	1.5	0.9	0.7	0.7	32, 0.7

halves k instead when more than 10% of the window’s block calls had their leading pair capped (drops are outrunning the support). Guard and fallback are untouched, so Theorem 6 applies with $k_{\max}$ in the drop bound. The rule is a prospective use of the model: a/b was fixed at 20 from the two pilot cells (gisette, ijcn1) before the other three were run, and $\hat{\eta}$ is estimated from the run’s own first window. On the three held-out cells the sizes it predicts from the first window (10 on w8a, 13 on a9a, 7 on covtype) are within one of the retrospectively fitted k^* (9, 13, 8), and the fixed sizes nearest to them are the best measured ones in Table 9. On the two pilot cells (Table 6) it settles at $k=9$ then 15 on gisette, tying the fixed sizes within CI, and at $k=7-8$ on ijcn1, where it is the fastest configuration (5.0 ± 0.5 s against 5.7 ± 1.9 s for $k=16$); the held-out cells’ isolated timings with the adaptive size are part of the final benchmark run. Table 11 tests it where η actually varies: synthetic L_2 instances ($n=2,000$ per class, $C=1$) whose transfer directions decorrelate as the dimension grows, so that $\hat{\eta}$ falls from 0.31 at $d=20$ to 0.001 at $d=1,000$. The rule tracks k^* across two orders of magnitude of $\hat{\eta}$ and matches the best fixed size within the resolution of these short single-seed runs (about 0.1 s) at every d .

7.5 Numerical check of Theorem 14

The theorem’s constants are computable on small instances. For 40 random hard-margin instances ($d = 4$, ten points per class, Gaussian clouds 2.5 apart) the SLSQP reference gives z^* , the faces, τ , $\alpha_{\min}$, $d_{\min}$ and σ (as the singular value of Assumption 13); 12 instances that fail strict complementarity or affine independence within 10^{-7} are skipped, and the remaining 28 are run with $k \in \{1, 4, 16\}$ to $\varepsilon = 10^{-10}$ with both supports snapshotted at every iteration (`src/ident_check.py`, 84 runs). Three things are checked. The weight inequality (F2), $\omega_t \leq h_t/\tau$, holds at every iteration of every run. From the first iteration at which the proof’s radius and weight conditions hold (the conditions on r_t , ω_t and $2h + R\sqrt{2h}$ that define H_{id} and H_{w}), claims (i) and (ii) hold to the end of every run: every removal of a non-face point is a drop, no non-face point enters, and after t_1 no index leaves. The third finding is that those conditions engage far later than the mechanism they certify. The last non-face support point leaves, and the last drop occurs, at $h \approx 10^{-3}$ (median; $h_0 \approx 6$); the proof’s conditions first hold at $h \approx 3 \times 10^{-13}$, and the sufficient h -thresholds $H_{\text{id}}, H_{\text{w}}$ lie at 3×10^{-18} . The theorem is therefore verified in its ingredients and contradicted nowhere, but its thresholds are pessimistic by some 14 orders of magnitude in h (about 7 in $\|z - z^*\|$) on these instances, most of it in the passage from r to h through $r^2 \leq (4R^2/\delta^*)\sqrt{2h}$. The large-data profiles of Figure 4 are consistent with the mechanism; they do not verify the thresholds, which no run of Section 6 reaches.

8 Related work

Distance duality and the TA (Kalantari, 2015; Gupta and Kalantari, 2026); MDM (Mitchell et al., 1974; López and Dorronsoro, 2015), Gilbert (Gilbert, 1966) and Wolfe (Wolfe, 1976); FW for polytope distance and SVMs (Gärtner and Jaggi, 2009; Clarkson, 2010; Tsang et al., 2005;

Ñanculef et al., 2014); the nearest-point and reduced-hull view (Keerthi et al., 2000; Bennett and Bredensteiner, 2000); away steps (Guélat and Marcotte, 1986), the pyramidal-width analysis (Lacoste-Julien and Jaggi, 2015) and the facial distance (Peña and Rodríguez, 2019). Boosted FW (Combettes and Pokutta, 2020), which also aggregates atoms into one direction, swap-step-free pairwise variants (Tsuji et al., 2022; Rinaldi and Zeffiro, 2023) and block-coordinate FW (Lacoste-Julien et al., 2013) are the closest algorithmic relatives; gap-safe screening (Fercq et al., 2015; Ndiaye et al., 2017; Shibagaki et al., 2016) and active-set identification (Bomze et al., 2020; Garber, 2020) the closest to Theorem 11, with (El Ghaoui et al., 2012; Ogawa et al., 2013; Wang et al., 2014) as its SVM ancestors (below).

Aggregated directions. Boosted FW (Combettes and Pokutta, 2020) builds its direction by greedily aligning a combination of atoms with the negative gradient, at several oracle calls per iteration, and inherits the FW rate through an alignment condition; our block fixes the pair coefficients γ_j from one scan and relies on the guard, not on alignment, for its rate.

Swap steps. Blended pairwise conditional gradients (Tsuji et al., 2022) also remove pairwise FW’s swap-step bound with a guard, there between the pairwise step and a FW step, and (Rinaldi and Zeffiro, 2023) avoid bad steps with a short-step chain. Our guard compares an aggregated block of k transfers with the single MDM step and falls back to an away step, so the drop-step count of away-step FW applies unchanged. Block-coordinate FW (Lacoste-Julien et al., 2013) blocks over *coordinates* of a product domain rather than aggregating pairs on one polytope.

Screening and identification. Gap-safe rules (Fercq et al., 2015; Ndiaye et al., 2017; Shibagaki et al., 2016) screen with a radius from the duality gap and identify the support in finite time; the active-set complexity of away-step FW (Bomze et al., 2020) and the strict-complementarity bounds of (Garber, 2020) give the same identification from the algorithm’s side. Theorem 11 is the polytope-distance instance, with constants tied to the TA certificate; the SVM rules of (El Ghaoui et al., 2012; Ogawa et al., 2013) and the exact data reduction of (Wang et al., 2014) are its ancestors.

Relation to other FW variants. The toward, away, pairwise and block rows of the step ablation in Section 6.1 are Frank–Wolfe, away-step FW, pairwise FW and our block step on the same polytope and objective; blended pairwise conditional gradients (BPCG) (Tsuji et al., 2022) is compared on real data in Table 7; Wolfe’s minimum-norm-point method (Wolfe, 1976) was not run. BPCG’s blending rule compares the away and FW gaps and its linear rate has the same pyramidal-width form as Theorem 6; our case (b) instead drops when the away step is clipped and otherwise compares the realised gains of the block, MDM, away and toward candidates, so each non-drop step is at least as good as the best single candidate, and it is the away fallback rather than the blending that yields the drop-step count. The facial distance of (Peña and Rodríguez, 2019) is a computable alternative to the pyramidal width; a bound on δ for the difference atom set $A = \{v_i - w_j\}$ in terms of the two class hulls is open.

Synthetic results in brief. Away and pairwise steps converge at $\varepsilon = 10^{-6}$ where the toward and midpoint steps of (Gupta and Kalantari, 2026) do not. Screening reduces 10,000 points to 36–1,054 survivors at $\varepsilon = 10^{-5}$. On seven synthetic and small-real problem classes (Figure 3, five trials each) the optimised solver is 1.1–1.2 $\times$ faster than LIBSVM on the four hard-margin classes (intervals overlap), ties LIBLINEAR on the synthetic L_2 margin, and is 1.7–2.7 $\times$ slower than NuSVC and LIBLINEAR on the ν -SVM and MNIST classes. On the four classes where single MDM also converges, blocks and guard change wall-clock by 0.9–1.2 $\times$; their clear wins are the two classes (tight tolerance, MNIST) where single MDM does not converge at all.

9 Conclusion

Guarded block transfers give the Triangle Algorithm a linear rate with explicit constants and no combinatorial swap-step bounds, and a safe screening rule with a finite activation bound. The experiments locate the boundary: sparse support and well-separated hulls, where the solver ties LIBSVM, beats the column-oracle solvers BPCG and kernel SMO (not LIBSVM with the kernel in memory), and supplies a certificate, against dense support or nearly touching hulls, where LIBLINEAR or LIBSVM is one to two orders of magnitude faster. Limitations: screening pays only when the certificate closes; the implementation is NumPy-level and dense; the constants are loose; and the ν -SVM variant is not analysed. Section 7 answers the question the workshop version left open: aggregation acts only on the per-iteration term of the work, by the exact factor Q/κ per step, drops are an identification cost that stops once the optimal faces are found, and the block size that pays most is $\sqrt{(a/b)(1-\eta)/\eta}$, which the solver can set from the κ it already computes; on column-bound problems no block size helps, and the paper says so.

Declarations

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Competing interests. [TODO: The author has no competing interests to declare, or state them.]

Data availability. The datasets analysed are the public LIBSVM benchmark sets a9a, w8a, ijcnn1, gisette and covtype (<https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/>); the synthetic instances are generated by the released code from fixed seeds.

Code availability. [TODO: public repository URL and archived release DOI (e.g. Zenodo)]; every table and figure is produced by a script named in Section 6.1.

Use of AI tools. AI tools were used in the preparation of the code, experiments and manuscript; the author verified all results and is responsible for the content.

Prior publication. A six-page version of this work was submitted to the OPT 2026 workshop (NeurIPS 2026, non-archival; [TODO: state the decision at submission time]); this article extends it with [TODO: list: full proofs, the exact reductions, the isolated timing study, the n-scaling experiment, ...].

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